



Measuring virtual embodiment: A psychometric investigation of a standardised questionnaire for the psychological sciences

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ABSTRACT

Psychological science increasingly employs virtual reality (VR) as a therapeutic and experimental tool. When represented by an avatar, a user's self-perception is altered through a full-body perceptual illusion known as virtual embodiment. However, inconsistent measurement of this construct derives uncertainties about the efficacy of VR-driven behavioural, cognitive, and social-psychological changes. Accordingly, we sought to provide psychological scientists a valid and reliable measure to assess the virtual embodiment construct during experimental and therapeutic simulations. In a repeated-measures experimental paradigm, participants ($N = 86$) completed a series of visuo-motor tasks in-vivo and in VR. Our study had three objectives: 1) compare visuo-motor performance across in-vivo and VR tasks; 2) assess the psychometric properties of theoretically foundational body ownership, agency, and self-location dimensions from a virtual embodiment questionnaire widely used in the literature; 3) analyse the simplified 11-item questionnaire's underlying structure using principal component analysis (PCA). We found that VR embodiment does not directly align with physical embodiment when considering proprioception and kinaesthesia, and there is limited support for convergent and discriminant validity of body ownership, agency, and self-location as independent dimensions of the embodiment construct. However, there is strong support for criterion validity when these dimensions comprise the overall measurement of embodiment. The PCA revealed a straightforward and theoretically interpretable underlying structure with three components: *embodiment*, *disembodiment*, and *agency*. This study offers psychometric support for the exploration of the simplified 11-item questionnaire as a theoretically grounded measure for behavioural scientists who seek to investigate virtual embodiment as a psychological construct.

1. Introduction

Psychological studies using immersive virtual reality (VR) are increasingly integrating an embodiment technique, in which users are represented by a virtual or artificial body (also known as the avatar). The *embodiment illusion* refers to the experience of being inside, having, and controlling the virtual body, inducing the *sense of embodiment* in VR, which emerges when properties of the virtual body are processed as if they belong to the user's physical body (Blanke & Metzinger, 2009; de Vignemont, 2011; Kilteni et al., 2012). Recent reviews have highlighted relationships between this experience of virtual embodiment and psychological outcomes (Cieslik et al., 2020; Furlan & Spagnoli, 2021; Geraets et al., 2021; Park et al., 2019; Tassinari et al., 2022). However, this body of work remains limited in its measurement of the embodiment construct, with some investigations implicitly assuming the influence of

embodiment without its dedicated measurement (Furlan & Spagnoli, 2021; Tassinari et al., 2022) and others relying on heterogeneous, non-standardised outcome measures (Forster et al., 2022; Mottelson et al., 2023). Further, although it is often expected that virtual simulations encapsulate or intimately mimic real-world phenomena (Parsons, 2015), the sense of embodiment toward the virtual body in VR appears to be reduced compared to the physical experience of being embodied (Döllinger et al., 2023).

Although several measures of virtual embodiment have been proposed (Dobricki & de la Rosa, 2013; Eubanks et al., 2021; Gonzalez-Franco & Peck, 2018; Roth & Latoschik, 2020), there is limited research investigating their psychometric properties, with validations being predominantly restricted to factor analytic approaches. Moreover, their constructions have deviated from a foundational theoretical structure (Longo et al., 2008), in which the sense of embodiment is experienced as

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a subjective sense of body ownership, agency, and self-location toward the virtual body (Kilteni et al., 2012, 2015). Currently concealed by ongoing measurement variability, however, is the influence of these foundational individual differences on VR-augmented person-centred practices. To encourage the uptake of a standardised, straightforward, and theoretically meaningful measure of virtual embodiment in the psychological sciences, we investigate the psychometric properties of the body ownership, agency, and self-location dimensions of Gonzalez-Franco and Peck's (2018) measure of virtual embodiment (hereafter referred to as the Embodiment Questionnaire [EQ]). Understanding the degree to which the embodied user perceives that the virtual body has functionally replaced their physical body is critical to understanding the profound effects we can observe in behaviour, cognition, or emotion during and after virtual immersions (Gall et al., 2021; Lindner, 2021; Wienrich et al., 2021).

1.1. Virtual reality, embodiment, and psychological science

The use of VR as a therapeutic and investigative tool has become prominent across diverse branches of psychological research and intervention. With the knowledge that virtual embodiment can change self-perception, cognition, attitudes, and behaviour (Slater & Sanchez-Vives, 2014), immersive simulations have been employed to induce emotional, psychological, and physical reactions that enable individuals "to acquire new learning for the benefit of their psychological well-being" (Cieslik et al., 2020). In social psychology, simulated interaction with members of different social groups facilitates intergroup contact (Alvidrez & Peña, 2020; Peña et al., 2021; Taylor et al., 2020), and embodying an alternative identity (e.g., race, culture, gender, age, class) can reduce bias, enhance social cognition, and increase empathy and prosocial behaviour (Farmer & Maister, 2017; Tassinari et al., 2022). Psychotherapy undertaken in VR can enhance the therapeutic alliance (Miloff et al., 2020; Wei et al., 2023), enable personal problem solving (Slater et al., 2019), improve mood (Bernardo et al., 2021), and increase self-compassion (Falconer et al., 2014). In clinical settings, virtual interventions have been conducted to assist cognitive-behavioural therapists in treatment for post-traumatic stress, anxiety, depression, psychosis, substance use, eating disorders, neurodevelopmental disorders, and pain management (Cieslik et al., 2020; Freeman et al., 2017; Geraets et al., 2021; Park et al., 2019). Virtual intervention is also emerging in the field of forensic psychology to assist rehabilitation related to sexual assault, domestic violence, and aggression (Seinfeld et al., 2018; Tuente et al., 2020).

The increasing availability of VR technologies has allowed mental health practitioners to expand the accessibility of their services through integration with online and self-led delivery methods, reducing barriers associated with remote travel and limited resources (e.g., cost, time, specialist personnel; Lindner et al., 2019). Compared to traditional approaches, clients report greater satisfaction with and acceptability of immersive interventions and treatments (Maples-Keller et al., 2017; Mazurek et al., 2019). However, recent systematic analyses have revealed mixed results when considering the efficacy of VR compared to gold-standard in-vivo procedures (Cieslik et al., 2020; Park et al., 2019), with some recommending VR as an alternative replacement therapy (Deng et al., 2019; Gonçalves et al., 2012; Maples-Keller et al., 2017) and others considering it only as an add-on (Botella et al., 2015; Oing & Prescott, 2018; Viana et al., 2020; Zeng et al., 2018). Although results vary across targeted psychological domains, specific phobias, eating disorders, and social interactions have received the greatest attention (e.g., Butler & Heimberg, 2020; Lemmer & Wagner, 2015; Wechsler et al., 2019). In particular, this research highlights the necessity for embodiment within a virtual avatar to enhance realism and efficacy of exposure, cognitive behavioural therapy, and perspective taking (Furlan & Spagnoli, 2021; Geraets et al., 2021; Tassinari et al., 2022). To clarify mixed outcomes when implementing VR procedures, practitioners and researchers alike may benefit from integrating a standardised measure of

virtual embodiment.

1.2. The embodiment construct: underlying structure and psychological processes

As a psychological construct, embodiment is not unique to VR. Embodiment has been defined across contexts as the sense of one's own body (Arzy et al., 2006), bodily self-consciousness (Legrand, 2006), corporeal awareness (Berlucchi & Aglioti, 1997), and a unique sense of somatic knowledge (Bermúdez, 1998). Embodiment theory suggests that psychological states arise from the physical states of the mind and body, and these bodily experiences allow us to understand ourselves and others (Barsalou et al., 2003; Niedenthal et al., 2005; Semin & Smith, 2002). Further, embodied cognition describes how perceptual and motor systems are fundamentally integrated with cognitive processing (Shapiro & Spaulding, 2021). Psychological processes then, whether occurring during interpersonal interactions or independently within an individual, cannot be fully understood in isolation from the body. While embodiment broadly encompasses complex relationships between the physical body and the mind, embodiment in the context of immersive virtual environments is determined by experiences of having a virtual body or through representations of the self as a virtual avatar. Importantly, the sense of embodiment in VR is induced by a full-body perceptual illusion, in which the view of the physical body and its surrounding environment are replaced with alternative visual stimuli (Kilteni et al., 2012, 2015). However, the aforementioned interplay between perception, action, and cognition remains active for the virtually embodied user.

Indeed, research has shown that psychological processes are not independent from a virtual body. Performance and behaviours across cognitive, social, behavioural, and perceptual-motor domains are hindered in the absence of a virtual body, but these improve when users are designated an avatar (Blanke & Metzinger, 2009; Herrera & Bailenson, 2021; Murray & Sixsmith, 1999). Thus, it appears that physical embodiment constituents, such as bodily self-consciousness, corporeal awareness, and embodied cognition, persist during virtual experiences (Herrera & Bailenson, 2021; Peck & Good, 2023). However, the success of attributing these phenomena to the virtual body (over the physical body) depends on the strength of the embodiment illusion (Kilteni et al., 2015; Peck & Good, 2023). Moreover, the degree to which individuals will experience this illusion is largely determined by three factors: body ownership, agency, and self-location (Kilteni et al., 2012; Longo et al., 2008). Originally identified in research using the rubber hand illusion—in which participants perceive an artificial limb as part of their own body—these factors emerged as measurable components associated with subjective experiences of altered or manipulated embodiment (Botvinick & Cohen, 1998; Gallagher, 2000). More recently, a breadth of research maintains that these factors are key to quantifying and investigating the psychological construct of embodiment (e.g., Beaufils & Berland, 2022; Eubanks et al., 2021; Tosi & Romano, 2023).

1.2.1. Visuo-motor synchrony for body ownership, agency, and self-location

Conceptualisations of body ownership, agency, and self-location are relatively consistent across the literature in cognitive, psychological, and computational approaches to virtual embodiment. Body ownership refers to the user's perception of owning the virtual body and perceiving the virtual avatar as the source of bodily sensations (de Vignemont, 2011; Tsakiris, 2010). Agency is defined as the conscious intention and performance of actions, including the sense of controlling one's own bodily movements (Spanlang et al., 2014; Tsakiris et al., 2006) and command over the actions of the avatar (Aoyagi et al., 2021). Finally, self-location is understood as one's perceived localisation within the spatial boundaries of a body during static and dynamic bodily states, including perceived origin of ego-centric visuospatial perspective (Ehrsson, 2007; Huang et al., 2017). That is, when the avatar is

co-located with the user's physical body and shown from a first-person perspective, users perceive that the self is located within the virtual body (Maselli & Slater, 2013; Petkova et al., 2011; Slater et al., 2010).

While these factors are traditionally evaluated via visuo-tactile stimulation, whether for a single body part (e.g., Romano et al., 2021) or full-body illusion (e.g., Tosi & Romano, 2023), embodiment is not a static experience (Giummarra et al., 2008) nor is visuo-tactile stimulation essential for the embodiment illusion (Slater et al., 2009, 2010). Drawing on sensorimotor theories of perception-action coupling (Covarrubias et al., 2017), these factors can be evaluated via any spatio-temporal congruency of cross-modal stimuli (e.g., visuo-motor, visuo-proprioceptive; Kilteni et al., 2015; Slater et al., 2008; Slater et al., 2010). Agency, in particular, operates in the cognitive domain of motor control (Zito et al., 2020), and full-body illusions using kinematic avatars afford the dynamic investigation of embodiment via visuo-motor tasks. Across experimental paradigms, both body ownership and agency are elicited by visuo-motor synchrony between the user's subjective experience of self-generated actions and the avatar's kinematic activity during motor task performance (e.g., González-Franco et al., 2010; Jeong et al., 2021; Odermatt et al., 2021; Sanchez-Vives et al., 2010). Notably, visuo-motor synchrony appears to be the only reliable manipulation to induce agency during virtual embodiment (Mottelson et al., 2023).

Like the physical body in the everyday world, the virtual body determines how users perceive and act within the virtual environment (Beaufils & Berland, 2022; Matamala-Gomez et al., 2021). However, users experience corporeal bilocation within virtual environments, such that their own body remains in the physical world while simultaneously navigating the virtual world from the avatar's perspective (Furlanetto et al., 2013; Gorisse et al., 2017). In particular, vestibular, proprioceptive, and kinaesthetic systems continue to be affected by natural physics, while visual and spatial position and orientation are determined by virtual stimuli (Beaufils & Berland, 2022). Thus, reliance on visuo-tactile stimulation alone may not capture the multidimensional complexities involved in the embodiment illusion when the kinematic body is taken into consideration.

1.2.2. Interdependence in the underlying structure

Although underpinning theory highlights body ownership, agency, and self-location as crucial determinants of the sense of embodiment, these components are not necessarily independent. Research has been unable to distinguish between these three dimensions in factor-analytic validation attempts (Peck & Gonzalez-Franco, 2021; Romano et al., 2021), noting that body ownership, in particular, might be comprised of agency and self-location (Braun et al., 2018; Kilteni et al., 2012; Tsakiris et al., 2006). Indeed, many studies investigating the sense of embodiment focus primarily on body ownership as a broad overarching construct with additional contributing factors (e.g., Kirsch & Kunde, 2023; Lee et al., 2021). Conversely, users may experience elevated agency over a virtual body, even when body ownership and self-location are experimentally reduced (Kim et al., 2020). These mixed results suggest that even if virtual embodiment may be best evaluated as a whole or interdependent construct, its comprehensive measurement should retain information regarding the three primary contributing factors.

Most recently, Tosi and Romano (2023) were able to distinguish between body ownership, agency, and self-location alongside an independent disembodiment factor by using exploratory graph analysis in favour of factor analysis. In this work, *disembodiment* describes the fading sensation or reduced awareness of the user's physical limbs during synchronous visuo-tactile stimulation of a static body in VR. Traditionally, embodiment and disembodiment are opposed, such that embodiment of the fake limb leads to displacement of the real limb (Longo et al., 2008; Romano et al., 2021). However, the sense of embodiment in VR can be induced without the concurrent displacement of the user's physical body (Tosi & Romano, 2023). That is, embodiment

and disembodiment are not mutually exclusive, but instead can be experienced simultaneously and to varying degrees. Notably, this understanding of disembodiment differs from that as reviewed by Mottelson et al. (2023), in which disembodiment refers to conditions of conflicting sensory information (e.g., incongruent cues, asynchronous stimulation) that are expected to contribute to detachment from the virtual body. In the context of interdependence, however, Tosi and Romano (2023) note the importance of avatar co-location to enable ownership and agency over the virtual body, while also allowing the physical body to co-exist.

Further potential for interdependence is indicated when considering the range of approaches employed to investigate these constructs. For example, visuo-tactile stimulation has been used to investigate body ownership, whereas visuo-motor stimulation addresses both body ownership and agency (Kokkinara & Slater, 2014; Tsakiris, 2010). Additionally, body ownership may be enhanced by the simple presentation of a virtual body (Eubanks et al., 2021; Maselli & Slater, 2013; Peck & Tutar, 2020), whereas visuo-motor synchrony is required for agency and self-location (Eubanks et al., 2021; Kokkinara & Slater, 2014; Odermatt et al., 2021; Tsakiris et al., 2006). Although the embodiment illusion is said to be robust to variations in the appearance of the virtual body with respect to identifying characteristics (e.g., age, race, gender; Gonzalez-Franco & Peck, 2018), perceived body ownership, agency, and self-location remain subjective and vary across individuals (Gonzalez-Franco et al., 2019; Peck & Good, 2023). Given the extensive range of cross-modal stimulation employed across methodologies, ongoing advances in VR technologies that are used to investigate virtual embodiment, and individual differences in the degree to which the sense of embodiment is experienced, a generalisable measure suited to the psychological sciences is needed.

1.3. Overview of existing measures

At least four approaches have been taken, to date, toward constructing and validating a standardised self-report measure of virtual embodiment. First, Dobricki and de la Rosa (2013) constructed a 27-item questionnaire measuring self-identification, agency, and spatial presence by adapting items from previous research on the rubber hand illusion (Longo et al., 2008), full-body illusions (Ionta et al., 2011; Lenggenhager et al., 2007; Petkova & Ehrsson, 2008), and spatial presence (Wirth et al., 2008). Self-generated items were additionally developed to assess bodily self-perception (Dobricki & de la Rosa, 2013). Although the final measure included items for body ownership and self-location within the self-identification dimension, research has shown that spatial presence (i.e., the relationship between self and environment) is not related to the sense of embodiment (Gonzalez-Franco & Peck, 2018; Kilteni et al., 2012; Roth & Latoschik, 2020) and does not require bodily representation (Kilteni et al., 2012). Moreover, experimental validation employed a third-person perspective visuo-tactile task, which limits the application of this measure under the necessary condition of avatar first-person perspective (Maselli & Slater, 2013).

Second, Roth et al. (2017) constructed a 13-item questionnaire measuring acceptance (i.e., body ownership), control (i.e., agency), and change in self-perception by extending a previous questionnaire on the illusion of virtual body ownership (Lugrin et al., 2015), with items adapted from previous research on the rubber hand illusion (Kalckert & Ehrsson, 2012), appearance (Maselli & Slater, 2013), visual perspective (Slater et al., 2010), virtual mirror use (González-Franco et al., 2010), and robot embodiment (Powers & Kiesler, 2006). Roth and Latoschik (2020) later extended and validated their original measure using data across four experiments investigating level of immersion, personalisation, behavioural realism, and latency, without the integration of a virtual mirror. The final 12-item Virtual Embodiment Questionnaire (VEQ) measures body ownership, agency, and change in body schema (i.e., weight, height, width, form/appearance). However, neither

investigation of the VEQ incorporated an assessment of self-location. Researchers noted a particular lack of available questionnaires from which to model self-location items (Roth & Latoschik, 2020) as well as a potential confound between self-location and presence (Roth et al., 2017).

Third, Gonzalez-Franco and Peck (2018) constructed a 25-item questionnaire with six sub-components measuring body ownership (EQ1-5), agency and motor control (EQ6-9), tactile sensations (EQ10-13), location of the body (EQ14-16), external appearance (EQ17-20), and response to external stimuli (EQ21-25). Items were adapted from 33 previous studies, conducted between 1998 and 2017, that used an embodiment illusion in VR and additionally cited the original rubber hand illusion paradigm (Botvinick & Cohen, 1998). The authors first classified all items originating from these studies into the six aforementioned categories, which were generated to describe the types of questions most frequently used based on experimental design. The final subset of 25 questions was then chosen based on frequency of use and experimental significance across the 33 reviewed studies (see Gonzalez-Franco and Peck's [2018] Table 1 for review).

Peck and Gonzalez-Franco (2021) later validated their original measure using data from nine previous experiments with 443 questionnaires that used at least part of the original 25-item measure. The final 16-item questionnaire re-distributes the six initial underlying constructs into appearance, ownership, response, and multisensory dimensions (Peck & Gonzalez-Franco, 2021). Although the multisensory dimension aggregates items for agency and self-location with tactile sensations, items in the multisensory (and ownership) subscales are simultaneously included across the other three subscales. Additionally, six of the theoretically salient items were excluded from this model (i.e., Body Ownership: EQ2, EQ3, EQ4, EQ5; Agency: EQ7, EQ9). While the oblique rotation employed to construct this final questionnaire allows for inter-relationships between subscales, its non-independent, interstitial structure and divergence from prominent theory may serve as a barrier for behavioural scientists who wish to quantify and investigate virtual embodiment as a psychological construct.

Fourth, Eubanks et al. (2021) proposed a short form of the embodiment questionnaire for use within virtual environments. Five items for body ownership ($n = 3$), agency ($n = 1$), and self-location ($n = 1$) were drawn from Gonzalez-Franco and Peck (2018), and five items were adapted from the Spatial Presence Experience Scale (Hartmann et al., 2016). The resulting 5-item Preliminary Embodiment Short Questionnaire (pSEQ), which includes one item for body ownership and two items each for agency and self-location, was validated using data across two experiments investigating avatar and motion tracking fidelity (Eubanks et al., 2021). Despite limitations with sample size, gender balance, and reduced sensitivity of the short form, the pSEQ explicitly measures all three components of the sense of embodiment. However, Eubanks et al. (2021) observed a requisite to create or adapt additional items, particularly for body ownership, to better capture the virtual embodiment construct. Thus, we do not further evaluate the pSEQ in the current study.

In sum, of the four previous approaches to constructing and

validating a self-report measure of virtual embodiment, only two have comprehensively included the three individual factors underlying the embodiment construct (i.e., body ownership, agency, self-location; Eubanks et al., 2021; Gonzalez-Franco & Peck, 2018). While Dobricki and de la Rosa (2013) included items addressing each factor, self-location was examined using third-person perspective, in which the sense of embodiment is either reduced or entirely absent (Goris et al., 2017; Maselli & Slater, 2013). Similarly, Roth and Latoschik (2020) did not address self-location. With the exception of the pSEQ, additional subscales across these questionnaires address constructs that may co-occur with the sense of embodiment (e.g., spatial presence, change in body schema, cross-modal stimulation), but which vary independently of the traditional embodiment factors (Roth & Latoschik, 2020). When considering that the sense of embodiment is experienced as a subjective sense of body ownership, agency, and self-location toward the virtual body (Kilteni et al., 2012), Gonzalez-Franco and Peck's (2018) original EQ appears the most comprehensive in aligning with these theoretical foundations.

1.4. The current study

This body of research has yet to investigate a breadth of psychometric properties required to support the use of a self-report questionnaire as an efficacious tool to measure the complex experience of embodiment in virtual environments. Accordingly, the objectives of the current study were three-fold. Given that virtual embodiment appears to be reduced compared to physical embodiment (Döllinger et al., 2023), we first investigated whether visuo-motor performance differs between procedures undertaken in the real environment and repeated in VR. Consistent with recent work (Döllinger et al., 2023; Mottelson et al., 2023), visuo-motor tasks were employed to allow for the examination of virtual embodiment under conditions of visuo-motor congruency. Second, we assessed convergent, discriminant, and criterion validity by administering body ownership (EQ1-5), agency (EQ6-9), and self-location (EQ14-15) items from the EQ against previously standardised measures for body connection, agency, proprioceptive acuity, and the rubber hand illusion. As the third self-location item (EQ16) is relevant only when the virtual body is not co-located with the participant (Gonzalez-Franco & Peck, 2018), we evaluated 11 of the original EQ items as applicable to co-located, first-person perspective avatars. Third, we analysed the structure of the 11-item EQ by employing principal component analysis with oblique rotation to allow correlation among components, which is theoretically and empirically aligned with previous work (Kilteni et al., 2012; Romano et al., 2021).

The overall aim of this work is to provide researchers and practitioners a valid, reliable, and generalisable measure to assess the psychological construct of virtual embodiment during experimental and therapeutic simulations. To do this, we specifically adhere to the underlying theoretical structure of the sense of embodiment identified in the cognitive and experimental literature. Thus, we do not evaluate Gonzalez-Franco and Peck's (2018) EQ dimensions for tactile sensations, external appearance, and response to external stimuli. Indeed, these have been identified as unessential to the sense of embodiment (Falcone et al., 2023) and vary in their relevance across specific and extended experimental manipulations, such as threat induction (Fribourg et al., 2021; González-Franco et al., 2014) or interacting with objects in the environment (Bhargava et al., 2023). For instance, the language employed for certain items from these subscales requires alterations that are dependent on specific experimental procedures (e.g., "I felt a ___ sensation in my body when I saw ___"). Similarly, the avatar's appearance is frequently treated as a modifiable factor to induce targeted social or behavioural change (e.g., Banakou et al., 2020; Peck et al., 2013). Although the appearance of the avatar is inherently present when visually representing users in VR, it remains an auxiliary sub-component of body ownership (Falcone et al., 2023; Kilteni et al., 2012).

Additionally, we adhere to the need for independence in the

Table 1
PAT performance in-vivo and in VR.

Trial Type	In-Vivo		VR		Z
	M (SD)	Median	M (SD)	Median	
Ipsilateral Dominant	11.52 (5.06)	10.90	17.05 (10.13)	14.20	−4.34**
Ipsilateral Nondominant	11.84 (5.46)	10.00	16.61 (8.03)	16.00	−4.18**
Contralateral Dominant	15.99 (9.60)	14.30	18.78 (9.62)	16.70	−2.13*
Contralateral Nondominant	15.66 (8.11)	13.20	17.03 (8.58)	15.00	−1.74

Note. * $p < 0.05$. ** $p < 0.001$.

underlying subscale structure of a self-report measure for use in psychological science, such that each subscale measures a homogenous construct, evaluates different aspects of virtual embodiment, and no items are repeated across subscales (Oosterveld et al., 2019; Rattray & Jones, 2007). Thus, we do not employ the questionnaire structure proposed by Peck and Gonzalez-Franco (2021) due to its non-independent, interstitial dimensionality. Instead, we sought to optimise item selection to reduce overlap among the components in the final measure.

2. Method

2.1. Participants and design

Undergraduate psychology students volunteered to participate in a study assessing differences in basic visuo-motor performance in-vivo and in VR. We employed a within-subjects experimental design, in which participants performed a series of three visuo-motor tasks, first in the real environment (i.e., in-vivo) and repeated in VR, followed by a battery of questionnaires. To determine the target sample size for this study, we conducted a priori power analyses for within-subjects correlations, paired samples t-tests, and multiple regression with two predictors. Using G*Power 3.1, alpha was set to the conventional value of 0.05 and power at 80%, which revealed a minimum of 82 participants would be needed to detect moderate effect sizes (e.g., $r = 0.3$, $d = 0.5$, $f^2 = 0.15$; Faul et al., 2007). This recruitment target also exceeded our minimum sample expectations for the principal component analysis, seeking at least five independent measurements for each of the 11 items sampled (Kline, 2014; MacCallum et al., 1999). Although 104 participants were recruited to complete the study, 18 cases were excluded from analysis due to incomplete data resulting from equipment failure (e.g., apparatus, software) or inability to complete study procedures.

The final sample ($N = 86$; $M_{age} = 20.95$, $SD = 6.12$) was comprised of 51 women, 33 men, one non-binary individual, and one individual who did not identify their gender. Participants were largely of Anglo-Australian (40.7%) and Asian descent (31.4%), followed by those from Middle Eastern (11.6%), European (10.5%), and mixed cultural backgrounds (3.5%). Additionally, one participant identified as Black, and one identified as Hispanic in self-describing their cultural and family backgrounds. More than half of the participants (65.1%) reported previous VR experience (limited: $n = 43$, moderate: $n = 10$; frequent: $n = 3$).

2.2. Visuo-motor tasks and standardised measures

The body of work surrounding embodiment illusions consistently finds that the sense of embodiment is enhanced by rendering avatars that are co-located with the user's physical body, viewed from a first-person perspective, and which map the user's self-generated movements to the virtual body (e.g., Banakou & Slater, 2014; Beaufils & Berland, 2022; Gonzalez-Franco & Peck, 2018; Kiltani et al., 2012; Kiltani et al., 2015; Kokkinara & Slater, 2014; Landau et al., 2020; Maselli & Slater, 2013; Slater et al., 2010). Although these conditions replicate the experience of being embodied in the real world, individuals report less body ownership and agency toward the virtual body compared to their own physical body (Döllinger et al., 2023). However, the question remains as to whether kinaesthetic and proprioceptive perception might be altered in VR. Namely, the interaction of top-down knowledge about being in an altered environment and the conflicting presentation of bottom-up stimuli from two bodily sources (i.e., physical, virtual) creates a condition that is only present during the VR experience. By holding the aforementioned conditions constant across three standardised visuo-motor tasks completed in the real environment and replicated in VR, we can assess differences in visuo-motor performance and self-reported bodily sensations. Visuo-motor tasks are illustrated in the following video: <https://youtu.be/9c5xGBVbOTc>.

2.2.1. Kinaesthetic and visual imagery questionnaire (KVIQ) – visuo-motor task 1

We administered the KVIQ, which quantifies visual and kinaesthetic dimensions of motor imagery, to measure whether participants' kinaesthetic sense differed between the real environment and the virtual environment. The KVIQ visuo-motor task requires that participants first observe the demonstration of a seated anatomical movement (i.e., neck flexion/extension, forward shoulder flexion, forward trunk flexion, knee extension, hip abduction, foot tapping, foot external rotation), before executing the movement once and then mentally visualising the movement without physical performance (Malouin et al., 2007). Participants then rate their visualisation on clarity of image (i.e., visual subscale) from 1 (*no image*) to 5 (*image as clear as seeing*) and intensity of sensation (i.e., kinaesthetic subscale) from 1 (*no sensation*) to 5 (*as intense as executing the action*). We administered seven of the 10 items from the short form of the KVIQ, excluding three items that were not reproducible in VR (i.e., shoulder elevation, elbow flexion/extension, thumb-fingers opposition). After observing, performing, and visualising an anatomical movement, participants verbally responded to both subscales and scores were recorded. Scores on each item were summed for the two subscales (visual, kinaesthetic), with higher scores indicating greater kinaesthetic awareness. The KVIQ has previously demonstrated construct validity, test-retest reliability, and good internal consistency for both subscales (visual: $\alpha = 0.89$, kinaesthetic: $\alpha = 0.87$; Malouin et al., 2007). In the current study, subscales demonstrated good to excellent internal consistency in the real environment (visual: $\alpha = 0.90$, kinaesthetic $\alpha = 0.88$) and in VR (visual: $\alpha = 0.92$, kinaesthetic: $\alpha = 0.89$).

2.2.2. Proprioceptive acuity test (PAT) – visuo-motor task 2

We administered the PAT, which quantifies the ability to judge the magnitude of joint rotation, to measure whether participants' proprioceptive sense in VR would be influenced by proprioceptive drift. Additionally, PAT performance in VR was nominated as a comparative measure with the EQ self-location subscale. Self-location is typically evaluated by comparing the third- and first-person perspectives of the virtual body, with focus on evoking the out of body experience (e.g., Ehrsson, 2007). However, displacement measures, such as induced proprioceptive drift, may be more useful in deriving information about perceived self-location when a first-person perspective is maintained (Botvinick & Cohen, 1998; Roth & Latoschik, 2020). To induce proprioceptive drift in VR, the virtual arm was offset 20 cm laterally relative to the physical arm position. Participants were able to view their virtual arm during target placement, but not during target replication, during the VR iteration of PAT procedures as described below. For the in-vivo iteration of the PAT, participants wore a blindfold throughout the task.

The PAT visuo-motor task requires that participants replicate five target forearm positions (150°, 120°, 90°, 60°, 30°; baseline = 0°) during ipsilateral and contralateral iterations while seated (Goble, 2010; Horváth et al., 2020). Participants completed 20 randomised trials, comprised of 10 dominant-handed and 10 non-dominant-handed trials, in the real environment and in VR for a total of 40 trials. For ipsilateral iterations, participants replicated the target indicated by the demonstrator using the same arm after being returned to baseline. For contralateral iterations, participants replicated the target indicated by the demonstrator using the opposite arm without a return to baseline. The difference between the target and replicated positions was recorded for each trial, and absolute values were averaged across the five trials for ipsilateral dominant, ipsilateral non-dominant, contralateral dominant, and contralateral non-dominant sets, with higher scores indicating lower accuracy. The PAT has previously demonstrated convergent and divergent validity, test-retest reliability, and acceptable internal consistency ($\alpha = 0.56$ –0.68; Horváth et al., 2020). In the current study, the PAT demonstrated similar internal consistencies in the real environment (ipsilateral: $\alpha = 0.57$ –0.66, contralateral: $\alpha = 0.52$ –0.66) and in VR (ipsilateral: $\alpha = 0.64$ –0.73, contralateral: $\alpha = 0.69$ –0.72). Reliability

metrics <0.60 for dominant-handed trials in the real environment indicate the potential for attenuation, or underestimation of the true effect size, for these trial types.

2.2.3. Free movement exercise – visuo-motor task 3

Consistent with previous VR movement-based research (Banakou et al., 2016), we choreographed a 1-min Tai Chi movement sequence with six basic movements to allow participants to experience full-body motor control over their virtual avatar. The sequence began with y-axis head rotation followed by arm raises (i.e., rising and sinking), before the more complex poses integrating the arms, legs, and torso (i.e., parting the wild horse's mane [right/left], repulse the monkey, embrace the moon). All poses were performed in the cardinal planes of anatomical motion (i.e., sagittal, coronal, transverse), without circular self-locomotion, to allow for novice levels of expertise. Participants completed the sequence in the real environment and in VR and were instructed to mirror the actions of the demonstrator.

2.3. Self-report measures

2.3.1. Embodiment questionnaire (EQ)

Three of the six subscales from the original EQ were included for validation in the current study. Participants responded to five items for body ownership (EQ1-5), four items for agency (EQ6-9), and two items for self-location (EQ14-15) on a Likert scale from 1 (*strongly disagree*) to 7 (*strongly agree*). The third self-location item (EQ16) was excluded, as this item only addresses instances in which the virtual body is not co-located with the participant (i.e., third-person perspective). The 11 items were presented to participants following the visuo-motor tasks undertaken in VR. As the virtual arm was offset 20 cm laterally relative to the physical arm position during the PAT, we additionally evaluated participant's sense of self-location directly following this task. Following the recommendations of Gonzalez-Franco and Peck (2018), three body ownership items (EQ2, EQ3, EQ5), one agency item (EQ9), and one location item (EQ15) were reverse scored. The total and three subscale scores were then calculated by averaging responses to each item, with higher scores indicating a greater sense of embodiment. Using this recommended scoring, the EQ demonstrated acceptable internal consistency overall ($\alpha = 0.62$, 95% CI [0.48, 0.73]). However, the subscales in their original format primarily demonstrated poor to questionable levels of internal consistency (body ownership: $\alpha = 0.53$, 95% CI [0.36, 0.67]; agency: $\alpha = 0.11$, 95% CI [-0.24, 0.38]; self-location: $\alpha = 0.11$, 95% CI [-0.37, 0.42]). The degree to which these levels of reliability align with previous work is unclear, as reliability metrics were not reported by Gonzalez-Franco and Peck (2018). Thus, we also examined the confidence intervals around Cronbach's alpha. While the total score and body ownership subscale reached moderate levels of reliability, the agency and self-location subscales demonstrated poor reliability (Koo & Li, 2016). Consequently, correlations using the agency and self-location subscales are likely to be attenuated.

2.3.2. Scale of body connection (SBC)

To our knowledge, there are no standardised self-report measures that independently quantify body ownership. As previous work has highlighted the relationship between body ownership and body awareness (Döllinger et al., 2022, 2023; Filippetti & Tsakiris, 2017), we nominated the SBC as a comparative measure of body ownership with the EQ body ownership subscale. The 20-item SBC quantifies general tendencies toward body awareness (i.e., sense of attention to the body and bodily sensations) and body dissociation (i.e., sense of separation from the body and avoidance of sensations; Price & Thompson, 2007). Participants responded to items on a 5-point Likert-type scale from 0 (*not at all*) to 4 (*all of the time*). Body dissociation items were reverse scored, and scores on each item were averaged for the two subscales, with higher scores indicating a greater general or global sense of body connection. The SBC has previously demonstrated construct validity and

acceptable to good internal consistency across cultures and languages ($\alpha = 0.63$ – 0.86 ; Price et al., 2017). In the current study, the SBC subscales demonstrated good internal consistency (awareness: $\alpha = 0.77$, dissociation: $\alpha = 0.79$).

2.3.3. Sense of agency rating scale (SOARS)

We nominated the SOARS as a comparative measure of the sense of agency with the EQ agency subscale. The SOARS quantifies state, or context specific, agency across two underlying factors. Involuntariness refers to the perceived degree of personal control over self-generated actions, while effortlessness refers to the perceived degree of passiveness and cognitive automaticity of self-generated actions (Polito et al., 2013, 2014). The SOARS is typically used in hypnosis research, in which individuals perceive reduced agency over self-generated actions. However, we can apply the SOARS in the current study with the understanding that while the visuo-motor tasks were designed for ease and simplicity (i.e., high effortlessness), participants in a non-hypnotic state experience an active degree of control over self-generated actions (i.e., low involuntariness). Participants responded to 10 items on a Likert scale from 1 (*strongly disagree*) to 7 (*strongly agree*). Scores on each item were summed for the two subscales, with higher scores indicating a greater degree of involuntary control and a lower degree of effort. The SOARS has previously demonstrated convergent and divergent validity, test-retest reliability, and good to excellent internal consistency for the two subscales ($\alpha = 0.73$ – 0.91 ; Polito et al., 2013). In the current study, the subscales demonstrated primarily acceptable internal consistency in the real environment (involuntariness: $\alpha = 0.67$, effortlessness: $\alpha = 0.76$) and in VR (involuntariness: $\alpha = 0.64$, effortlessness: $\alpha = 0.61$).

2.3.4. Sense of agency scale (SOAS)

We included the SOAS measure of trait, or global, sense of agency to evaluate participants' general, context-free beliefs of core agency (Tapal et al., 2017). Participants responded to 13 items on a Likert scale from 1 (*strongly disagree*) to 7 (*strongly agree*). Scores on each item were summed for a total score, with higher scores indicating a greater sense of trait agency. The SOAS has previously demonstrated convergent, divergent, and incremental validity, test-retest reliability, and sufficient internal consistency ($\omega = 0.75$ – 0.80 ; Tapal et al., 2017). In the current study, the SOAS demonstrated good internal consistency (in-vivo: $\alpha = 0.82$, VR: $\alpha = 0.83$).

2.3.5. Rubber hand illusion questionnaire (RHIQ)

To assess criterion validity, we nominated the RHIQ as a comparative measure of the overall sense of embodiment with the EQ total score. The RHIQ quantifies perceived sense of ownership when one's physical body is replaced artificially (Romano et al., 2021). For use in VR procedures, the phrase 'rubber hand' was modified to 'virtual hand'. The 16 items comprising the embodiment (i.e., increased sensation or ownership toward the artificial body) and disembodiment (i.e., decreased sensation or ownership toward the physical body) subscales were included in the current study. As there was no physical touch manipulation in our experimental procedures, the two items measuring physical sensations were excluded. Participants responded to items on a 7-point Likert scale from 1 (*strongly disagree*) to 7 (*strongly agree*). The total and subscale scores were calculated by averaging responses to each item, with higher scores indicating greater perceived sense of embodiment. The RHIQ has previously demonstrated good to excellent internal consistency ($\alpha = 0.80$ – 0.94 ; Romano et al., 2021). In the current study, the RHIQ also demonstrated good internal consistency (total: $\alpha = 0.84$, embodiment: $\alpha = 0.87$, disembodiment: $\alpha = 0.71$).

2.4. Materials and apparatus

2.4.1. Virtual environment, room setup, and motion tracking

The cross-platform gaming engine Unity-3D (version 2019.4 [LTS] Unity Technologies, San Francisco, USA) was used to create the virtual

environment, which was designed to represent a modern waiting room (Fig. 1) and exercise room (Fig. 2). The waiting room scene featured a floor-to-ceiling virtual mirror, in which participants could view their avatar and self-generated movements. The exercise room scene featured a room-length virtual mirror as well as a table and chairs for seated exercises.

Participants used the HTC Vive Pro Eye system with two hand controllers to interact with the virtual environment. Additionally, HTC Vive 2.0 and 3.0 trackers (6-DoF) were attached to the participant and demonstrator to allow for full-body, real-time motion control of the virtual avatar. The headset, controllers, and trackers were mapped to the participant and the demonstrator based on their positions in 3D space (e.g., head, chest, wrists, ankles). Three Vive 2.0 base stations were mounted to the anterior, posterior, and right perpendicular walls to trace position and orientation of these tracking components. The three visuo-motor tasks were performed on the diagonal of the room, such that the participant and demonstrator faced each other, to enable continuous motion tracking at 100 Hz without occluding the base stations. For seated exercises, two chairs were placed on the diagonal and a portable table was placed in the centre. Fig. 3 displays the setup of the lab in which participants performed the in-vivo exercises. The room's dimensions were 417 cm $\times$ 533 cm, and the virtual exercise room was scaled to these dimensions using the Vive's room scale setup function to trace the boundaries of the play area.

2.4.2. Virtual avatars

Three virtual avatars (two feminine, one masculine) were created using Adobe Fuse and Blender 3D modelling software (see Fig. 4). Avatar models were designed to minimise clipping and the inaccurate intersection of the virtual body parts. Inverse kinematics and physics were then applied to the avatars in Unity to further minimise the potential for these effects. Avatars were rigged for animation using Mixamo (<http://www.mixamo.com/>) and controlled in real time by the participant and demonstrator via the Vive tracking components. To calibrate the avatars to the user's physical body, height and arm length (in centimetres) were entered into the experimenter's Unity character user interface. Avatars were scaled to these values by referencing an assigned offset value for each tracking component placed at the head, chest, wrists, and ankles, accounting for variations in rotation and head-to-chest, arm-to-shoulder, and waist-to-ankle ratios. Although the specifics of avatar acceptance are beyond the scope of this paper, indications of avatar acceptance by user ethnicity are summarised in Appendix A.

2.4.3. Technical specifications and system performance

The virtual environment and HTC Vive components were hosted in the lab on a desktop computer with a Windows 10 Enterprise LTSC operating system. The system's processor (CPU) was an 11th Generation Intel Core i7-11700 with a clock speed of 2.5 GHz and 48 GB of installed

RAM. The graphics card (GPU) was an Nvidia GeForce RTX 3070. The monitor was an HP X27q with a resolution of 2560 $\times$ 1440, 8-bit colour depth, RGB colour format, Standard Dynamic Range (SDR) colour space, and a 164.834 Hz refresh rate. The HTC Vive system's display refresh rate was 90 Hz, field of view 110°, and combined resolution 2880 $\times$ 1600 pixels (1440 $\times$ 1600 pixels per eye). The Vive base stations had a positional accuracy of 0.5 cm and a latency of 11.11 ms. The Vive tracker's 6-DoF position (x, y, z) and angle (pitch, yaw, roll) data were updated at a fixed rate of 50 Hz, with the maximum latency between tracker and avatar movement <40 ms.

2.5. Procedure

Fig. 5 displays an overview of the experimental procedure. Participants were guided through the experimental tasks by the demonstrator, while a second experimenter ran the virtual components on the lab's desktop computer. Before undertaking the visuo-motor tasks, participants were asked to wear the Vive ankle and chest trackers and were provided the Vive controllers for wrist tracking. While directly facing the demonstrator in the real environment, participants completed the KVIQ, PAT, and free movement exercises. Standardised instructions for each task were given verbally by the demonstrator. Following the in-vivo tasks, participants completed the SOARS (i.e., state agency) with reference to the physical body and the SOAS (i.e., trait agency). Prior to entering VR, participants were asked to remain seated while fitting the HTC Vive headset. Participants then entered the virtual environment, first arriving in the virtual waiting room. Here, participants were instructed to stand up directly from their seated position, and the experimenter calibrated both the participant and demonstrator avatars standing in a neutral position (see Fig. 1). After calibration, the experimenter placed the participant and demonstrator avatars in the virtual exercise room (see Fig. 2) to complete the VR iterations of the KVIQ, PAT, and free movement exercises. Prior to beginning the first exercise, the demonstrator instructed participants to return to their seated position by sitting directly down in the real chair, which was co-located with the virtual chair behind them. Procedural reminders were provided in VR via graphical user interface, such that participants indicated to the experimenter when they were ready to continue to the next exercise by pointing their controller at the instruction panel and using the controller's trigger to click 'Next' (see Fig. 6). The experimenter then initiated each exercise to cue its associated virtual components (e.g., chair, table) via the experimenter user interface (see Fig. 7), and the standardised verbal instructions were again provided by the demonstrator. Procedures for the three visuo-motor tasks were directly replicated in VR, with the addition of the EQ self-location subscale measured in VR after the PAT for assessment of proprioceptive drift (see Fig. 8). After exiting VR, participants first completed the EQ, RHIQ, and SOARS (i.e., state measures) with reference to the virtual body followed by the SOAS and SBC (i.e., trait measures). All participants provided informed consent before undertaking study procedures as approved by the university's Human Research Ethics Committee. As experimental procedures took 50–60 min to complete, of which 15–20 min were spent in VR, participants received 1 h of course credit in exchange for taking part in the study.

3. Results

3.1. Task performance in-vivo vs VR

We conducted a series of repeated-measures tests to examine whether kinaesthetic sense, sense of agency, and proprioceptive acuity differed between the in-vivo and VR tasks. Cohen's *d* is reported as a measure of effect size, with 0.2, 0.5, and 0.8 reflecting small, moderate, and large effects, respectively (Cohen, 1988). Prior to conducting repeated-measures tests, outcomes of interest were examined for adherence to normality. The KVIQ, SOARS, and SOAS met conditions for parametric testing. As the PAT exhibited substantial negative skew, we



Fig. 1. Virtual Waiting Room. Participants first arrived in the waiting room for avatar calibration.

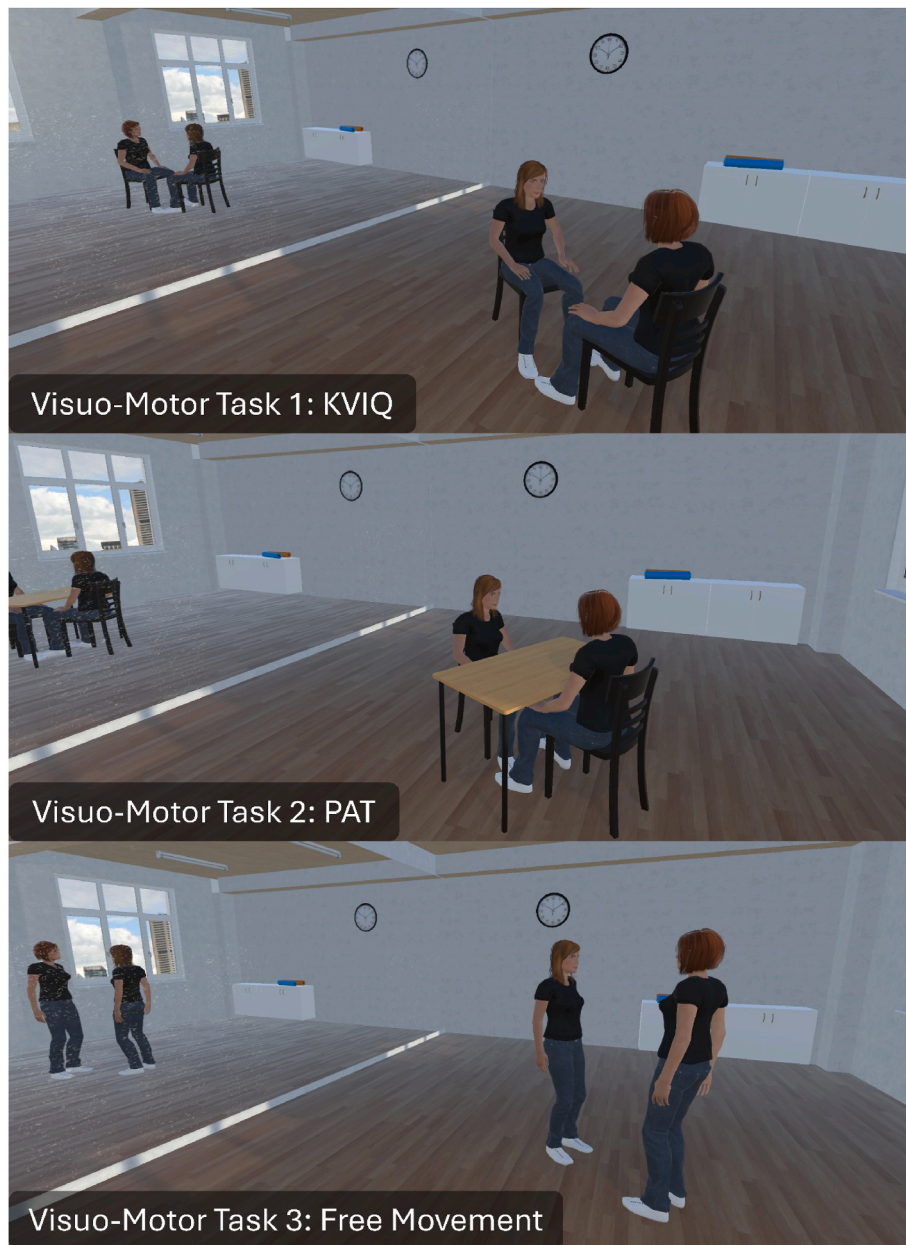


Fig. 2. Virtual Exercise Room. All visuo-motor tasks were undertaken in the exercise room. The participant's avatar faces the virtual mirror, while the demonstrator's avatar faces the participant.

employed non-parametric tests to analyse these data.

3.1.1. *Kinaesthetic sense*

Participants reported significantly higher scores on the KVIQ visual subscale in the real environment, $t(85) = -2.19, p = 0.031, d = 0.24$, such that the clarity of the image was reduced in VR compared to in-vivo. Conversely, participants reported significantly higher scores on the KVIQ kinaesthetic subscale in VR, $t(85) = 3.23, p = 0.002, d = 0.35$, such that the intensity of the sensations was greater in VR compared to in-vivo.

3.1.2. *Sense of agency*

For state agency, no significant differences were observed between in-vivo and VR tasks on the SOARS involuntariness, $t(85) = 0.07, p = 0.945, d = 0.01$, and effortlessness subscales, $t(85) = 0.12, p = 0.903, d = 0.01$. Additionally, SOAS scores for trait agency did not significantly differ, $t(85) = -0.58, p = 0.700, d = 0.06$. Taken together, the size of

these effects suggests that participants' perceived global and context-specific sense of agency remained consistent between in-vivo and VR experiences.

3.1.3. *Proprioceptive acuity*

Table 1 summarises participant performance outcomes for the PAT. Due to substantial negative skew, we used Wilcoxon signed rank tests to compare in-vivo and VR performance. Results indicate that participants' proprioceptive acuity was significantly reduced in VR, except for the contralateral non-dominant trials. The overall higher scores in VR indicate some influence of proprioceptive drift, such that participants were less accurate when the virtual arm was drifted laterally, particularly during ipsilateral trials. Despite this outcome, participants did not report significant differences in perceived self-location during this task compared to the overall VR experience, $t(84) = 1.94, p = 0.056, d = 0.21$. That is, perceived self-location was marginally lower during the PAT task specifically ($M = 4.10, SD = 1.02$), compared to perceived self-

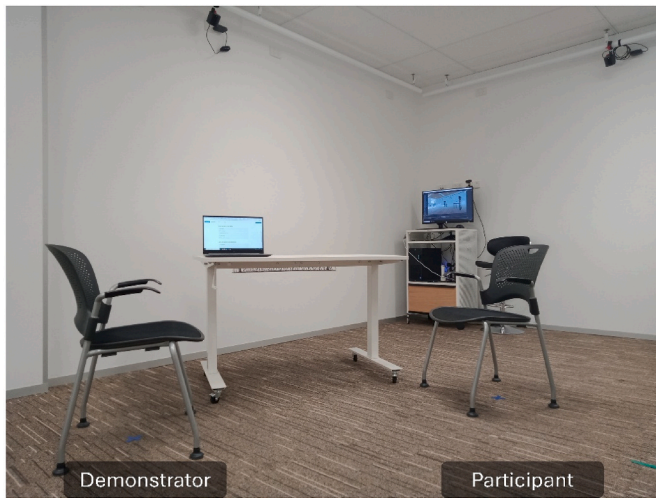


Fig. 3. Lab Setup. The location of the participant's chair in the lab was determined by matching its location in the virtual environment. Markers were placed on the floor to ensure the chairs and table were set consistently across participants and to ensure the co-location of the participant's chair when immersed in the virtual environment.

location when considered as a whole across the experimental tasks undertaken in VR ($M = 4.38$, $SD = 1.27$).

3.2. Validity testing

We assessed convergent and discriminant validity using a series of zero-order correlations, as summarised in Table 2. Convergent validity is indicated by moderate-to-high and significant correlations between a given EQ subscale and its nominated construct of comparison, whereas discriminant validity is indicated by little-to-no and non-significant correlations between the same EQ subscale and the remaining comparative constructs (Campbell & Fiske, 1959). Spearman's rho is reported for comparisons assessing the PAT to account for negative skew. Coefficients of 0.1, 0.3, and 0.5 reflect small, moderate, and large effect sizes (Cohen, 1988).

3.2.1. Body ownership and SBC

The SBC was theorised to align with the body ownership subscale of the EQ. Although negligible correlation was found with the SBC body awareness subscale, there was a small, positive correlation between body ownership and SBC body dissociation. This latter effect offers some evidence for convergent validity. Given that SBC dissociation is reverse

scored, users who report fewer tendencies toward avoidance of body awareness also experience more ownership of the virtual body. Regarding discriminant validity, body ownership showed minimal correlation, as expected, with the SOARS involuntariness subscale and all PAT trial types. However, there was an unexpected small-to-moderate, positive correlation with SOARS effortlessness.

3.2.2. Agency and SOARS

The SOARS was theorised to align with the agency subscale of the EQ. However, there were negligible correlations between agency and both SOARS subscales. Like body ownership, agency showed minimal correlation with all PAT trial types. However, there was no support for discriminant validity with the SBC, with agency demonstrating moderate, positive correlations with both SBC awareness and dissociation.

3.2.3. Self-location and PAT

Proprioceptive acuity in VR was theorised to align with the self-location subscale of the EQ. Here, we evaluate the self-location subscale as administered at the conclusion of the experimental tasks undertaken in VR. Negligible correlations were found between EQ self-location and PAT performance. There was some support for discriminant validity, as self-location demonstrated minimal correlation with SBC awareness, SBC dissociation, SOARS involuntariness, and SOARS effortlessness. However, it is important to note that the poor reliability of the EQ self-location subscale may have contributed to attenuation or bias toward zero across these results.

3.2.4. Relationships between body ownership, agency, and self-location

Given mixed results from the preceding validity testing and previous work highlighting how agency and self-location may comprise body ownership (e.g., Braun et al., 2018), we used a multiple regression analysis to assess the unique contributions of agency and self-location to body ownership. The overall model predicted 28.63% of the variance (adjusted R^2), $F(2, 83) = 18.05$, $p < 0.001$. Both agency, $b = 0.41$, $t(83) = 3.29$, $p = 0.001$, and self-location, $b = 0.35$, $t(83) = 4.51$, $p < 0.001$, independently predicted body ownership, accounting for 11.56% and 19.71% of the variance, respectively. Fig. 9 displays these results. Notably, agency and self-location were not significantly related ($r = -0.03$, $p = 0.389$), controlling for body ownership.

3.3. Total and RHIQ

The RHIQ total score was theorised to align with the EQ total score, and a moderate, positive correlation between these variables demonstrated good support for criterion (concurrent) validity ($r = 0.36$, $p < 0.001$). To further assess the unique contributions of the RHIQ subscales

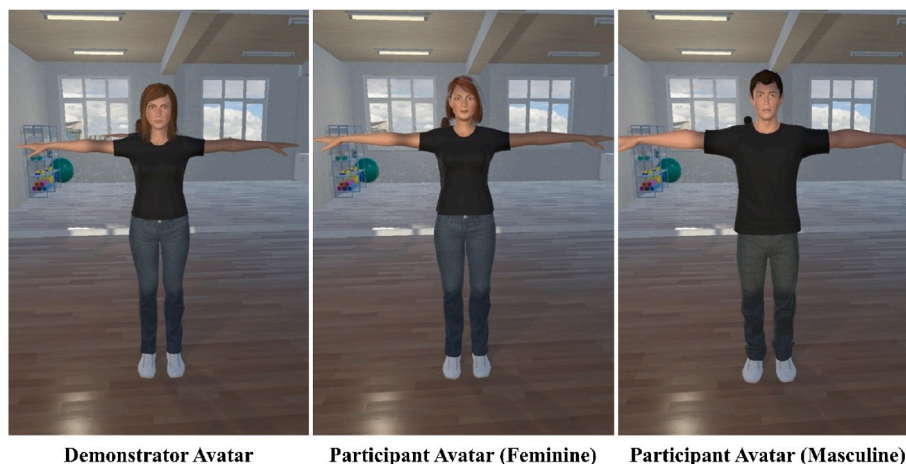


Fig. 4. Demonstrator and Participant Avatars. Participants were asked to choose the avatar that best represented their self-reported gender identity.

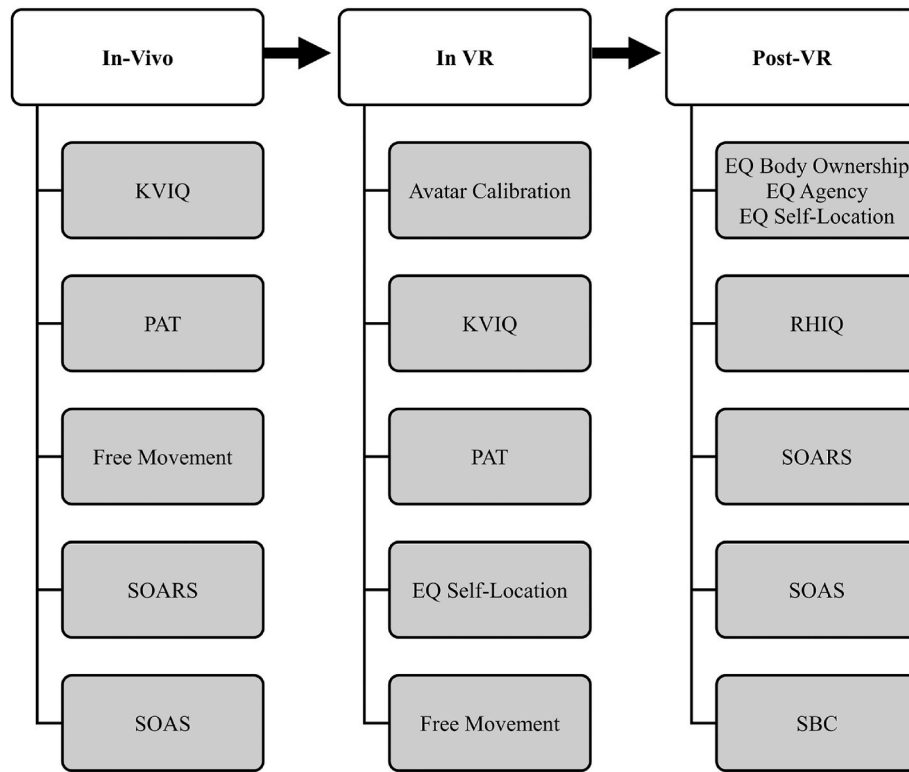


Fig. 5. Procedure Overview. Participants first completed the in-vivo portion of the procedure, followed by the VR exercises and post-VR questionnaires.



Fig. 6. Participant User Interface. Participants interacted with the instruction panel by pointing their controller and clicking the 'Next' button using the trigger located at the underside of the Vive controller.

to the EQ total, we employed a multiple linear regression with RHIQ embodiment and disembodiment as predictors. The overall model predicted 47.20% of the variance (adjusted R^2), $F(2, 83) = 38.99$, $p < 0.001$. Both RHIQ embodiment, $b = 0.49$, $t(83) = 8.47$, $p < 0.001$, and RHIQ disembodiment, $b = -0.32$, $t(83) = -5.08$, $p < 0.001$, independently predicted EQ total, accounting for 46.38% and 23.72% of the variance, respectively. Fig. 10 displays these results.

3.4. Principal component analysis

Overall, our initial validity testing yielded mixed results, and the EQ in its original format demonstrates generally poor reliability. Additionally, agency and self-location do not appear to be independent from

body ownership. Taken together, these results reinforce the need to re-evaluate the underlying structure of the EQ.

First, we explored the correlation matrix for the 11 EQ items (see Appendix B) to determine if items were sufficiently correlated with each other and whether any items demonstrated multicollinearity. Item seven demonstrated moderate-to-large correlations with two items and small-to-moderate correlations with another two items. The 10 remaining items each demonstrated moderate-to-large correlations (e.g., varying around 0.30–0.50; Cohen, 1988) with at least four other items, and no correlation coefficients exceeded 0.80 (Berry & Feldman, 1985; Hair et al., 2019; Vatcheva et al., 2016). Sampling adequacy and data suitability were assessed using the Kaiser-Meyer-Olkin (KMO) and Bartlett's sphericity measures, respectively. The KMO test demonstrated

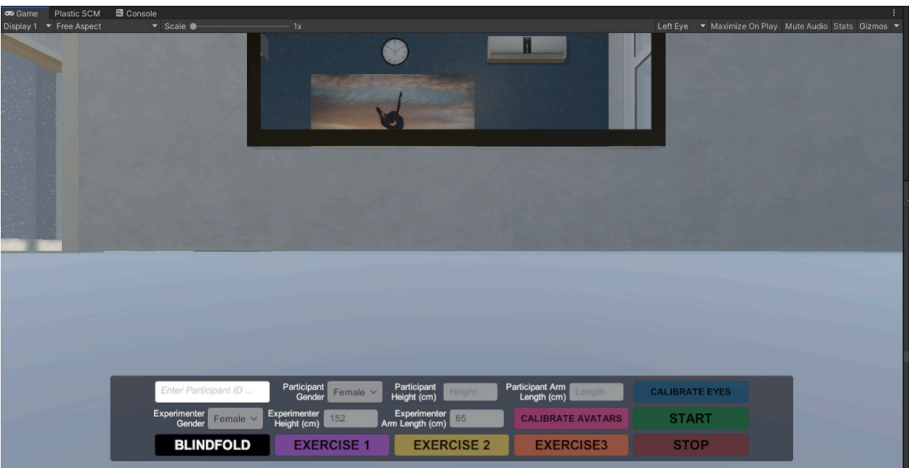


Fig. 7. Experimenter User Interface. The user interface allowed the experimenter to initiate each stage of the virtual procedure on the desktop computer (e.g., Start, Calibrate Avatars, Exercise 1, Exercise 2, Blindfold, Exercise 3).

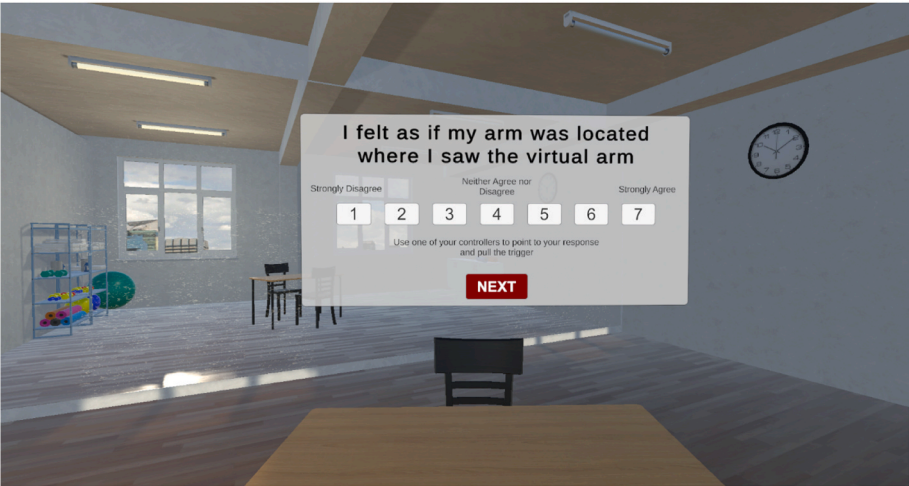


Fig. 8. EQ Location Administered in VR. Participants recorded their responses from 1 (*strongly disagree*) to 7 (*strongly agree*) by pointing their controller and using the Vive trigger to highlight and select their level of agreement with each statement.

Table 2
Zero-order correlations assessing convergent and discriminant validity.

	Body Ownership	Agency	Self-Location
SBC Awareness	−0.04	0.28*	0.02
SBC Dissociation	0.22*	0.29*	0.15
SOARS Involuntariness	−0.09	−0.15	−0.14
SOARS Effortlessness	0.24*	0.18	0.20
PAT Ipsilateral Dominant	0.01	−0.01	0.06
PAT Ipsilateral Nondominant	0.06	0.08	0.08
PAT Contralateral Dominant	−0.20	−0.17	−0.11
PAT Contralateral Nondominant	−0.04	−0.10	−0.07

Note. * $p < 0.05$. Spearman's rho is reported for PAT data. SOARS and PAT are evaluated as administered following VR visuo-motor tasks.

acceptable sampling adequacy at 0.71 for the overall dataset, with each individual item exceeding the 0.50 cut-off for inclusion (see [Appendix C](#); [Kaiser, 1974](#)). Thus, all 11 items were retained, and Bartlett's test of sphericity indicated the strength of the observed correlation matrix was suitable for principal component analysis, $X^2(55, N = 86) = 303.66, p < 0.001$ ([Bartlett, 1950](#)).

Items were entered into the PCA in their original format (i.e., unreversed), to evaluate the necessity of reverse scoring originally suggested by [Gonzalez-Franco and Peck \(2018\)](#) as a consistency check. Although

many of these items were removed from [Peck and Gonzalez-Franco's \(2021\)](#) analysis—citing low correlation, low KMO scores, the need for a virtual mirror, and complications with dual-balanced phrasing—EQ15 was retained in its unreversed format. Overall, it is not clear whether these items serve as consistency checks, dual-balanced statements, and/or require reverse scoring.

The PCA extracted three components with eigenvalues greater than one, explaining 63.16% of the total variance in the data. The average communality was acceptable for the current sample size at 0.63 (see [Appendix D](#); [MacCallum et al., 1999](#)), and 61% of non-redundant residuals exceeded an absolute value of 0.05. [Table 3](#) summarises unrotated item loadings, eigenvalues, and explained variance for the extracted components. Loading significance was determined based the current sample size by following the recommendations of [Hair et al. \(2019\)](#), suggesting unrotated item loadings should be examined for cross-loading at a threshold of $|0.60|$. We further examined cross-loadings in the unrotated component matrix by assessing whether the differences between primary and alternative component loadings were greater than 0.02 ([Costello & Osborne, 2005](#); [Howard, 2016](#)). Prior to rotation, one item loaded similarly onto two components, while an additional four items demonstrated differences less than 0.02 between the primary and alternative components.

To minimise cross-loadings, the component matrix was then rotated

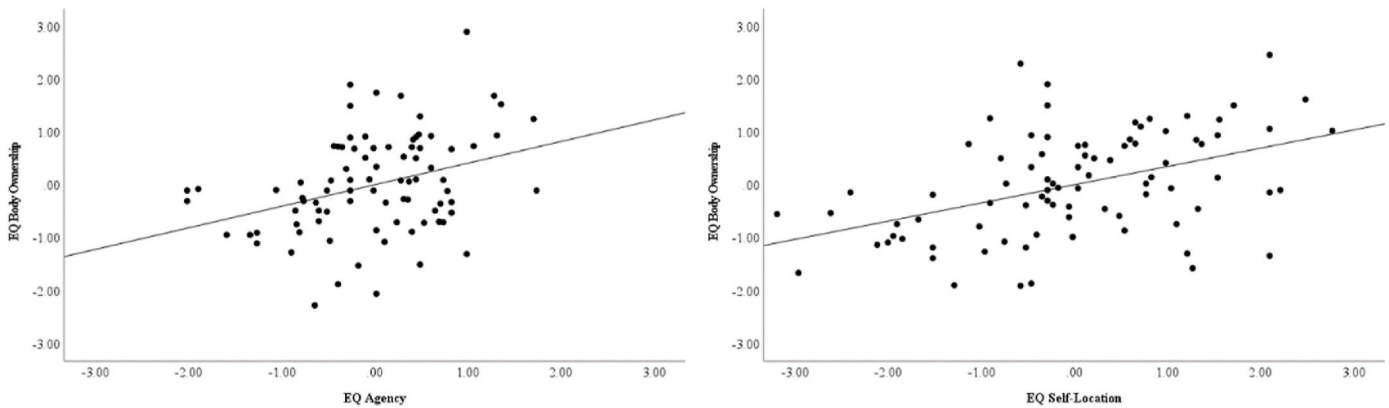


Fig. 9. Agency and self-location predicting body ownership.

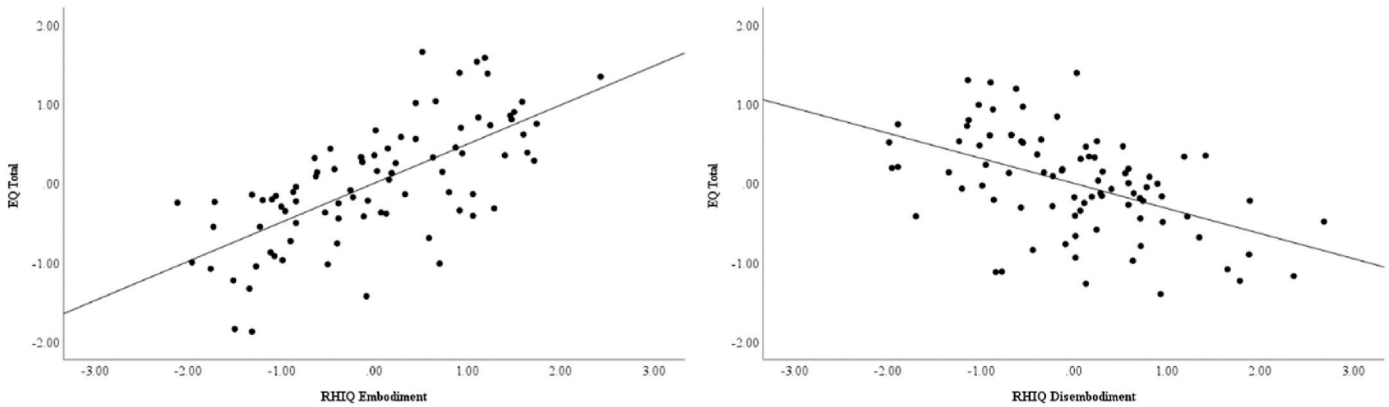


Fig. 10. RHIQ embodiment and disembodiment predicting EQ total (original scoring).

Table 3
Individual item loadings for the unrotated three-component model.

Item	Question	Component		
		1	2	3
EQ_1	I felt as if the virtual body was my body	0.62	0.61	-0.03
EQ_2	It felt as if the virtual body I saw was someone else	-0.64	0.46	0.22
EQ_3	It seemed as if I might have more than one body	-0.06	0.68	-0.02
EQ_4	I felt as if the virtual body I saw when looking in the mirror was my own body	0.67	0.53	-0.14
EQ_5	I felt as if the virtual body I saw when looking at myself in the mirror was another person	-0.60	0.17	0.57
EQ_6	It felt like I could control the virtual body as if it was my own body	0.68	0.19	0.51
EQ_7	The movements of the virtual body were caused by my movements	0.50	-0.13	0.74
EQ_8	I felt as if the movements of the virtual body were influencing my own movements	-0.16	0.60	0.09
EQ_9	I felt as if the virtual body was moving by itself	-0.38	0.64	-0.08
EQ_14	I felt as if my body was located where I saw the virtual body	0.61	0.32	-0.24
EQ_15	I felt out of my body	-0.33	0.67	-0.02
Eigenvalues		2.97	2.71	1.27
Percent Variance		26.99	24.68	11.50

Note. Values in bold indicate unrotated cross-loadings at values greater than or equal to |0.60|.

obliquely using Direct Oblimin rotation with Kaiser normalisation to allow for correlation between the components, and items were grouped according to component loadings greater than |0.60| in the rotated structure matrix (see Table 4; Hair et al., 2019). Although selecting components to retain is often based on the criterion of eigenvalues greater than one, we used multiple complimentary indices to determine the optimal number of principal components. While examination of the eigenvalues suggested retaining three components, an examination of the scree plot suggested a clear gap after the second component and a smaller gap after the fourth component (see Appendix E). To better

Table 4
Final structure matrix after oblique rotation.

Item	Embodiment	Disembodiment	Agency
EQ_1	0.78	0.31	0.39
EQ_4	0.83	0.21	0.32
EQ_5	-0.65	0.45	0.16
EQ_14	0.72	0.02	0.17
EQ_2	-0.43	0.69	-0.12
EQ_3	0.23	0.64	0.04
EQ_8	0.08	0.62	0.06
EQ_9	-0.02	0.72	-0.20
EQ_15	0.01	0.74	-0.12
EQ_6	0.45	-0.04	0.82
EQ_7	0.08	-0.24	0.87
Reliability	$\alpha = 0.75$	$\alpha = 0.72$	$\alpha = 0.69$
Component Correlations			
Embodiment	-		
Disembodiment	0.005	-	
Agency	0.189	-0.012	-

Note. Values in bold are greater than or equal to |0.60|. The negative sign for EQ_5 indicates the item should be reverse scored.

understand the elbow points in the scree plot, we employed two additional tests. Parallel analysis suggested extracting three components, whereas Velicer's Minimum Average Partial test suggested extracting two components (O'Connor, 2000). Given these conflicting results, we explicitly examined model fit without the third component, which is comprised of only two items. Here, the two-component solution resulted in greater residuals at 66%, less explained variance at 57.65%, and overall scale reliability dropped to 0.643. As excluding the third component reduced model fit and scale reliability, this component was retained as an independent factor for the final solution.

Although the third component exhibits somewhat lower internal consistency, this is not uncommon for a subscale with two items (Knekt et al., 2019). Indeed, residuals exceeded 50% for both solutions. However, the final three-component solution is appropriate for use with smaller sample sizes with the cut-off for significant component loadings set at a threshold of 0.60 (Hair et al., 2019), a 5:1 ratio of measurement to items factored (Kline, 2014; MacCallum et al., 1999), a good average communality, and an overall scale reliability of 0.691. Following the recommendations of Hair et al. (2019), only loadings ≥ 0.60 are significant for our sample size. Thus, each item loaded exclusively onto only one dimension, and no substantial cross-loadings were present in the final rotated model.

The first component captures items related to virtual body ownership and sensations attributed to the virtual body, which we propose to name *embodiment*. We propose to name the second component *disembodiment*, such that disembodiment denotes the degree to which users experience a sense of separation from their physical body when immersed in VR (Tosi & Romano, 2023). Finally, the third component is loaded by two items related to agency and motor control, which we propose to name *agency*. Table 4 summarises the three item clusters, categorised as Embodiment (Component 1), Disembodiment (Component 2), and Agency (Component 3), which exhibit acceptable-to-moderate reliability and minimal inter-scale correlations (Tavakol & Dennick, 2011).

It is necessary to evaluate the phrasing of items captured by the second component in relation to the working definition of disembodiment as separation from the physical body. This component captures two items related to the sensation of having two bodies (EQ2, EQ3), aligning with the notion that the user's physical and virtual bodies co-exist and not the assumption that the physical body is displaced by the virtual one (Tosi & Romano, 2023). Similarly, the second component captures EQ8, which more specifically describes the sensation of the virtual body influencing the user's physical body, again aligning with physical and virtual body co-existence. EQ15 directly aligns with the definition of disembodiment and is the only item that explicitly targets detachment from the physical body. EQ9, however, appears to target detachment from the virtual body, such that the user is not in control of the virtual body's movements. Although, removal of this item from the component reduced the dimension's internal consistency to $\alpha = 0.65$. Thus, it is possible that the disembodiment component is multidimensional, capturing different facets of disembodiment.

3.4.1. Reassessing convergent and discriminant validity

Component 1: Embodiment. The new Embodiment subscale demonstrated a large, positive correlation with RHIQ embodiment, $r(84) = 0.69, p < 0.001$, and negligible correlation with RHIQ disembodiment, $r(84) = 0.14, p = 0.189$. These results indicate strong evidence for convergent and discriminant validity for the Embodiment subscale.

Component 2: Disembodiment. The new Disembodiment subscale demonstrated a large, positive correlation with RHIQ disembodiment, $r(84) = 0.71, p < 0.001$, and negligible correlation with RHIQ embodiment, $r(84) = 0.10, p = 0.341$. Like Embodiment, the Disembodiment subscale demonstrates both convergent and discriminant validity.

Component 3: Agency. The new Agency subscale demonstrated a moderate, positive correlation with SOARS effortlessness, $r(84) = 0.30, p = 0.004$, and a small-to-moderate, negative correlation with SOARS

involuntariness, $r(84) = -0.24, p = 0.025$. As we designed the movement tasks to be high in effortlessness and low in involuntariness, the direction of these relationships is expected and offers good support for the Agency subscale as an independent component with two items. As we observed that the SBC was initially correlated with agency in tests for discriminant validity, we further tested the associations between the SBC and the new Agency subscale. Notably, Agency remained moderately and positively correlated with SBC awareness ($r = 0.25, p = 0.023$) and dissociation ($r = 0.25, p = 0.022$).

3.4.2. Reassessing criterion validity

Due to differences in the original scoring of the EQ (i.e., five reverse scored items) compared to the scoring structure revealed by the PCA (i.e., one reverse scored item), we re-evaluated the relations between the RHIQ total score and the EQ total score. Although the earlier analysis indicated some support for concurrent validity, we found a large, positive correlation ($r = 0.74, p < 0.001$) and much stronger support for concurrent validity using the new scoring. As before, we used a multiple linear regression to further understand the unique contributions of the RHIQ embodiment and disembodiment subscales. Here, the overall model predicted a greater proportion of the variance (adjusted R^2) at 54.78%, $F(2, 83) = 52.50, p < 0.001$. Again, both RHIQ embodiment, $b = 0.35, t(83) = 6.41, p < 0.001$, and RHIQ disembodiment, $b = 0.32, t(83) = 5.52, p < 0.001$, independently predicted EQ total, accounting for 33.18% and 26.83% of the variance, respectively. Fig. 11 displays these results. Notably, disembodiment positively predicted EQ total, whereas this relationship was negative when the original scoring was applied. That is, the initial negative relationship would indicate that as separation from the user's physical body increases (i.e., RHIQ disembodiment), the sense of virtual embodiment decreases. Instead, we now observe that as RHIQ disembodiment increases, the sense of virtual embodiment also increases.

4. Discussion

Embodiment techniques are increasingly being employed in psychological experiments and therapeutic interventions undertaken within VR (Furlan & Spagnoli, 2021). Although the user's perceived experience of embodiment in these procedures can impact social, emotional, cognitive, and behavioural outcomes, the virtual embodiment construct has been measured infrequently and inconsistently. To provide professionals in the psychological and behavioural sciences with a valid, reliable, and generalisable measure of virtual embodiment, we evaluated the psychometric properties of the EQ's body ownership, agency, and self-location dimensions (Gonzalez-Franco & Peck, 2018). Our results suggest that embodiment in VR does not directly align with physical embodiment when considering proprioceptive and kinaesthetic perception, and there is limited support for convergent and discriminant validity of body ownership, agency, and self-location as independent dimensions of the embodiment construct. However, there is strong support for criterion validity when embodiment is taken as a whole and comprised of body ownership, agency, and self-location items. Moreover, the *embodiment*, *disembodiment*, and *agency* dimensions generated by the principal component analysis demonstrated strong support for convergent and discriminant validity with existing measures of these constructs. Thus, we suggest that psychological scientists explore the use of the simplified 11-item measure as a generalised tool to quantify participants' perceived sense of embodiment during therapeutic interventions and experimental investigations undertaken within VR.

4.1. Considerations for virtual embodiment and visuo-motor perception

Embodiment in a virtual avatar allows individuals to perceive and interact with immersive virtual environments in a realistic manner (Kyrilitsias & Michael-Grigoriou, 2022). Our results, however, suggest that virtual embodiment should not be conceptualised as simply a direct

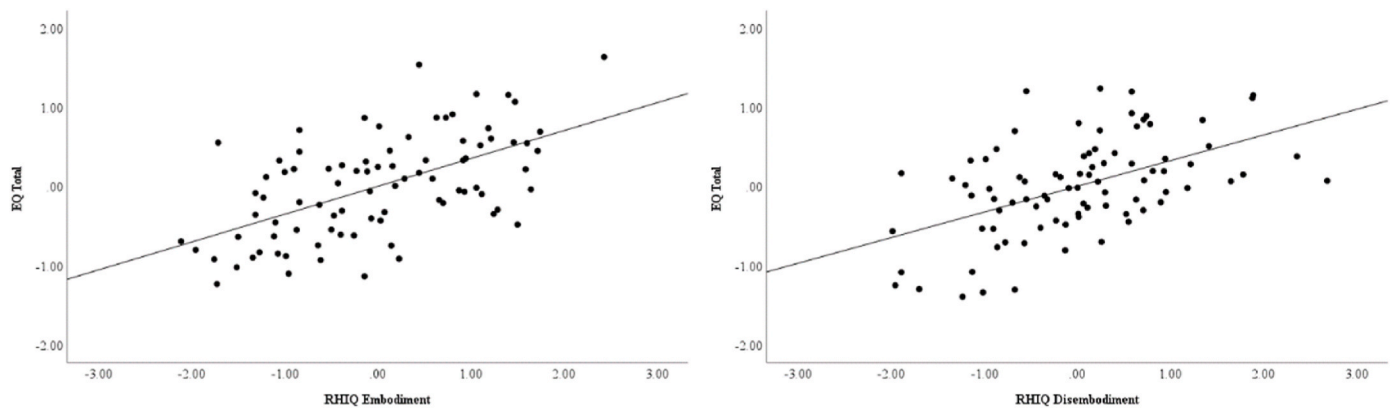


Fig. 11. RHIQ embodiment and disembodiment predicting EQ total (modified scoring).

translation from the physical body to the virtual body. Although participants' perceived sense of agency remained stable across state and trait levels, we observed changes to kinaesthetic and proprioceptive awareness. Previous research investigating visual-kinaesthetic experiences in VR attribute reduced presence and task performance to an incongruence between visual stimuli and bodily sensations (Bogicevic et al., 2019; Iachini et al., 2019). Consistent with this previous work, visual imagery on the KVIQ was reduced in VR compared to the real environment. Thus, altering the user's visual input may reduce their ability to accurately envision bodily self-representations due to conflict between degree of familiarity with the appearance of their physical body versus the virtual body.

Conversely, kinaesthetic imagery with respect to movement intensity increased in VR. Visual-kinaesthetic illusions can alter kinaesthetic perception through exaggerated visual input (Dufour et al., 2014; Eidenberger & Mossel, 2015; Meehan et al., 2002) or progressive de-coupling of vestibular and visually apparent motion cues (Tennent et al., 2017). During the KVIQ procedure, however, we maintained direct coupling of physical and apparent motion cues between the participant and their avatar. Thus, we expect that repeating the KVIQ within-subjects led to greater recall and a practice effect. Consistent with Beaufils and Berland (2022), kinaesthetic perception in VR appears to remain situated with the physical body, despite alterations to visual perception. However, as kinaesthetic imagery increased rather than remaining constant, future work should seek to administer this procedure between-subjects to elucidate whether VR indeed increases one's ability to envision kinaesthetic sensations.

Further consistent with previous work (Spitzley & Karduna, 2022; Tosi et al., 2022), we found that participants' sense of proprioception was vulnerable to drift. Although the effect for the contralateral non-dominant arm was not significant, proprioceptive acuity was consistently less accurate in VR when the virtual arm was shifted laterally. Of particular interest here are the ipsilateral trials, during which participants exhibited greater inaccuracies. Whereas contralateral trials maintain a reference point (i.e., the opposite arm remains in the target position), ipsilateral trials rely solely on memory to replicate the target position (Goble, 2010). Unlike kinaesthesia, proprioception in VR does not appear to remain situated with the physical body, and altered visual stimuli can override the sense of proprioception during task performance under conditions of visuo-proprioceptive incongruency (Spitzley & Karduna, 2022). However, as participants did not report any differences in perceived self-location during this task, visuo-proprioceptive incongruency does not appear to influence this dimension of embodiment (Bourdin et al., 2019; Caola et al., 2018; Lanillos et al., 2021).

In sum, we observed that tasks relying on mental imagery can be influenced by proprioceptive and kinaesthetic changes induced by virtual immersion. Mental imagery in the absence of external stimuli

operates during information processing, cognition, and learning, for example (Hofer et al., 2012; Kosslyn & Ochsner, 1994). Thus, psychological scientists should be aware of the reliance on visual input and the influence of illusory visual stimuli during therapeutic or experimental procedures that seek to apply virtual outcomes across to the physical world. Noting that the plausibility of virtual experiences depends on behavioural congruency between visually apparent movements and the user's actions (Berthiaume et al., 2018; Latoschik & Wienrich, 2022; Zopf et al., 2018), we recommend maintaining visuo-kinaesthetic and visuo-proprioceptive congruency to ensure the uptake of psychosocial, cognitive-behavioural, and motor-based VR interventions. Although state agency exhibited continuity across in-vivo and VR tasks, physical embodiment did not directly align with virtual embodiment with respect to proprioceptive and kinaesthetic factors. Thus, we reiterate the need to integrate measures of virtual embodiment to determine the robustness of the perceptual shift toward the virtual body across varying experimental, therapeutic, and technical standards.

4.2. Measuring the virtual embodiment construct: validity and reliability

Our initial investigation of convergent and discriminant validity yielded mixed results in attempting to establish body ownership, agency, and self-location as independent dimensions of virtual embodiment. Although previous research suggests that displacement measures (e.g., proprioceptive drift) may impact the experience of self-location (Dobricki & de la Rosa, 2013; Piryankova et al., 2014), we instead found that visuo-motor task performance does not correlate with self-location under conditions of proprioceptive drift. However, participant self-reports may differ from behavioural performance (Horváth et al., 2020), which could potentially offer some explanation of these results. While we found some positive relationships for body ownership and agency with previously established measures of related constructs, the mixed effects for discriminant comparisons and small convergent effect sizes are not sufficient to establish validity for employing these dimensions as subscales. Moreover, regression analyses revealed that both agency and self-location independently predicted body ownership, supporting the assumption that body ownership is comprised of the sense of agency and experience of self-location (Braun et al., 2018; Kiltner et al., 2012; Tsakiris et al., 2006).

Although previous conceptualisations of body ownership, agency, and self-location do not serve well as independent dimensions of virtual embodiment, these components remain essential to the perceived embodiment experience (Beaufils & Berland, 2022; Eubanks et al., 2021; Romano et al., 2021; Tosi & Romano, 2023). When we took embodiment as a whole construct comprised of body ownership, agency, and self-location items, there was strong support for convergent and criterion validity with the RHIQ. Both the embodiment and disembodiment dimensions of the RHIQ independently predicted virtual embodiment,

demonstrating concurrent validity. Additionally, the PCA revealed a similar underlying structure to the RHIQ, with independent dimensions suited to embodiment and disembodiment factors as well as a 2-item agency subscale. By retaining items essential to conceptualising embodiment, this structure remains theoretically meaningful and demonstrates content validity alongside psychometric reliability. Taken together, these observations encourage future investigation of the 11-item questionnaire in psychological science. To enhance generalisability across future endeavours, researchers and practitioners could work toward applying this selection of items among representative community, clinical, and cross-cultural samples when employing a virtual embodiment technique in psychological interventions.

4.3. Theoretical and practical implications

Embodiment has traditionally been investigated using the rubber hand illusion paradigm, however, this classical approach to experimentation is limited by visuo-tactile manipulation of a single body part (Botvinick & Cohen, 1998). Extending this approach, full-body illusions have allowed researchers to investigate the complex and multidimensional experience of embodiment to a greater extent. While this approach has allowed for a more comprehensive investigation of the sense of embodiment over the last 15 years, variation in measurement techniques is widespread (Forster et al., 2022; Furlan & Spagnoli, 2021; Mottelson et al., 2023; Tassinari et al., 2022). Given the dynamic nature of social interaction (Condie & Condie, 2021; Skvoretz & Fararo, 2016), self-other perception (Kahl & Kopp, 2018; McGann & De Jaegher, 2009), and embodied cognition (Wallace, 2021), measurement challenges are unsurprising among endeavours to quantify the role of the body in these domains. However, classical literature has already established a gold-standard measure of subjective and perceived embodiment (Botvinick & Cohen, 1998; Longo et al., 2008; Romano et al., 2021). Although most, if not all, measures of virtual embodiment draw on this literature (see Mottelson et al., 2023 for review), the current study appears to be the first to directly and empirically compare the two domains. By returning to the breadth of research on the rubber hand illusion, we can better integrate a standard by which to measure the virtual embodiment phenomenon. Indeed, the current study finds that a theoretically driven measure of virtual embodiment mirrors the underlying structure of the RHIQ (Romano et al., 2021), drawing together the somewhat disparate bodies of work situated in virtual embodiment and the classical psychological sciences.

Extending the work of Gonzalez-Franco and Peck (2018), the simplified 11-item measure demonstrates good construct (i.e., convergent, discriminant), criterion, and content validity and is psychometrically sound with a straightforward underlying structure consisting of three independent subscales. The first and second components mirror those of Romano et al. (2021), measuring embodiment and disembodiment for the virtual body. Like the RHIQ, embodiment and disembodiment are inversely related, but not mutually exclusive, such that the probability of attributing phenomena to the virtual body increases while the probability of attributing phenomena to the physical body decreases (Barnsley et al., 2011; della Gatta et al., 2016; Giummarra et al., 2008; Moseley et al., 2008; Romano et al., 2021). That is, while the physical body and the virtual body can be represented simultaneously in the user's cognitive self-representation, the likelihood of perceiving stimuli with reference to the virtual body over the physical body increases during a successful embodiment illusion (Folegatti et al., 2012; Giummarra et al., 2008; Romano et al., 2021; Tosi & Romano, 2023). As proprioceptive and kinaesthetic systems remain situated in the physical world, but their sensory perception can be altered in VR, assessing perceived embodiment and disembodiment independently could reveal the degree to which individuals might experience conflict between their two bodies during a given experimental or therapeutic task.

In sum, users will experience varying degrees of both embodiment and disembodiment, as the virtual body does not displace the user's

physical body (Tosi & Romano, 2023). However, measuring disembodiment in the context of full-body illusions is still emerging. The current study extends the work of Tosi and Romano (2023), who evaluated a static body in VR, by evaluating a virtual body in motion. We observed that disembodiment as measured by the RHIQ positively predicts the sense of embodiment as measured by the 11-item EQ, independently explaining 26.83% of the variance. Yet, the RHIQ's disembodiment items consistently refer to the user's physical body, whereas items loading onto the EQ's disembodiment component are more varied (e.g., sensation of multiple bodies, the virtual body moving autonomously). Consequently, EQ disembodiment may be multidimensional, capturing a degree of separation from the user's physical body as well as some detachment from the virtual body. Like embodiment, disembodiment has often been presumed to result from experimental cues, such as asynchronous stimulation, rather than being explicitly measured (Mottelson et al., 2023). Further iterations are required to understand whether additional disembodiment items could resolve potential multidimensionality. Thus, a comprehensive comparison of the EQ with the dimensions of other embodiment questionnaires (e.g., Dobricki & de la Rosa, 2013; Eubanks et al., 2021; Roth & Latoschik, 2020) should be addressed in future work.

The third component reflected context-specific sense of agency. Although this subscale is narrow with only two items, agency as an independent dimension contributed to scale reliability and demonstrated good construct validity with a gold-standard measure of agency (Polito et al., 2013). Consistent with recent work, users may experience a high degree of agency over a virtual body even when ownership and self-location are reduced (Jung et al., 2022; Kim et al., 2020). As agency indeed contributes to the success of embodiment illusions (Braun et al., 2018; Longo et al., 2008; Tosi & Romano, 2023), an independent examination of this subscale may offer researchers greater insight into the degree of perceived agency during procedures that require movement or motor control. Under conditions of synchronous visuo-motor activity, context-specific agency should be expected to increase (Abdulkarim et al., 2023; Buchholz et al., 2019). However, the opposite effect should be expected under conditions of asynchronicity or incongruence, for example during interventions for phantom limb pain (Dunn et al., 2017) or motor control disturbances (Aoyagi et al., 2021). Notably, body awareness remained associated with agency when examined against the final model, although it was initially theorised to align with body ownership. With previous work suggesting that assessing body awareness should be essential when evaluating virtual embodiment in the context of mental health (Döllinger et al., 2023), this result presents a compelling avenue for further exploration.

4.4. Limitations and future work

The current validation extends previous work in the measurement of virtual embodiment by assessing a theoretically driven measure, which includes 11 of the original items for body ownership, agency, and self-location described by Gonzalez-Franco and Peck (2018). We find that this measure demonstrates content, construct, and criterion validity for an underlying structure that is straightforward and theoretically interpretable along the dimensions of embodiment, disembodiment, and agency. However, the following limitations should be considered.

First, the 11-item measure was validated using a convenience sample of undergraduate students, with a mean age of approximately 20 years. However, previous research has found an age effect for virtual embodiment when comparing individuals over and under 30 years of age. Additionally, experimental research suggests that certain factors of embodiment may develop at different rates for varying age groups (Peck & Good, 2023; Weijs et al., 2021). Although this measure will be well-suited to experiments in psychological science that recruit university students and young adults for their studies, researchers should be mindful of potential age effects, for instance when recruiting community samples for clinical investigations. Future investigations should seek to

examine the sensitivity of this measure in discerning demographic differences across diverse and underrepresented populations in VR research.

Similarly, our avatars were designed to be representative of a majority in the Australian student sample (e.g., light skin tone, dark eye and hair colour). While avatars designed from White- and Asian-appearing models sought to represent these respective populations, users of colour were not well-represented by the avatars employed in the current study. Indeed, racial identification with avatars can influence embodiment. Greater discrepancies between an avatar's attributes and the user's physical attributes contribute to a reduced sense of embodiment in social VR (DeVeaux et al., 2023). Given that skin tone, hair texture, and dimensions of the eyes, nose, and mouth are salient in representing racial identity in virtual environments (Passmore & Mandryk, 2018), future work should aim to diversify avatar representation by facilitating participant choice from a wider selection of avatars or by enabling avatar customisation. Such an approach may not only increase embodiment across users with diverse identities, but also it may enhance avatar acceptance. The inclusion of measures such as the Player Identification Scale (Van Looy et al., 2012) or indices of humanness, attractiveness, and eeriness to measure the uncanny valley effect (Ho & MacDorman, 2017) could offer greater clarity on the relations between avatar acceptance, self-representation, and the sense of embodiment in future work.

Third, the reliability within each subscale, as measured by Cronbach's alpha, is somewhat lower than that demonstrated by Peck and Gonzalez-Franco (2021). Additionally, the internal consistencies for the SOARS and RHIQ were lower than those demonstrated in their original validation studies (Polito et al., 2013; Romano et al., 2021). Given that the SOARS was standardised to measure agency during hypnotic states, lower internal consistency in the current context is not overly surprising. However, there are currently no alternative self-report measures for the sense of agency, as the body of work in this area faces challenges in heterogeneous conceptualisations of agency (see Polito et al., 2013 for a review). For the RHIQ, we expect that our internal consistency values were lower than those demonstrated by Romano et al. (2021) due to the change in phrasing from 'rubber hand' to 'virtual hand' as well as the exclusion of two items measuring physical touch manipulation. However, the RHIQ performed well in the current context with high internal consistency for the total score and both subscales.

In comparing our subscale results to those of Peck and Gonzalez-Franco (2021), we expect our internal consistency values reflect the number of items and number of participants sampled, as well as the non-repetition of items across subscales. Whereas the current study evaluated 11 items, with components optimised to reduce overlap, among 86 participants ($\alpha = 0.69\text{--}0.75$), Peck and Gonzalez-Franco (2021) evaluated 16 items, with 10 items included in at least two subscales, among 443 participants ($\alpha = 0.72\text{--}0.82$). Indeed, both studies reflect acceptable-to-moderate levels of internal consistency and offer an improvement on the original structure and scoring as described in Gonzalez-Franco and Peck (2018), which achieved poor reliability in the current study. Comparing the disparate structures derived from each of these studies may provide an interesting line of inquiry for future investigation, in particular among previously standardised procedures frequently employed in the psychological and behavioural sciences.

Fourth, in line with our objective to support a theoretically driven and straightforward measure of virtual embodiment, we did not evaluate the additional three dimensions proposed by Gonzalez-Franco and Peck (2018). Although factors such as external appearance, tactile sensations, and response to external stimuli may be unessential to the sense of embodiment (Falcone et al., 2023), inclusion of additional items from these other subscales could enhance the application of the measure for specific experimental tasks involving tactile and reactionary manipulations. Although such research has been conducted independently (Bhargava et al., 2023; Fribourg et al., 2021; González-Franco et al., 2014), future work may wish to examine the impact on validity and

reliability when including these factors as independent dimensions. Additionally, some research has found that avatar appearance may impact the user's sense of ownership or presence in the environment (Ogawa et al., 2020; Pyasik et al., 2020), despite the robustness of the embodiment illusion to altered avatar appearance under conditions of motor control and first-person perspective (Fribourg et al., 2020). As we rendered avatars that did not explicitly manipulate physical appearance factors, any influence of external appearance was held constant in the current investigation. Thus, the measure is expected to perform well when arbitrary rather than personalised or manipulated representations of the avatar are implemented. However, virtual embodiment has been associated with modulations to self-esteem and empathy, particularly under conditions of social interaction and social identity manipulation (Herrera & Bailenson, 2021; Tassinari et al., 2022). Thus, our future work will look at the importance of external appearance as it relates to empathy across various social contexts.

4.5. Conclusion

The body and mind share an intimate connection. Given the increasingly ubiquitous augmentation of person-centred practices within VR, it is essential to consider the degree to which psychological processes become linked with the virtual body. In this paper, we introduce a versatile tool aimed at promoting the increased measurement of virtual embodiment among psychological scientists across investigative and therapeutic domains. As embodiment in virtual worlds does not directly replicate embodiment in the physical world, it is necessary to account for virtual embodiment as a unique construct that can vary across individuals. By employing a theoretically meaningful evaluation as part of our measurement criteria, we can better extend the knowledge of full-body illusions in VR and their effects on behavioural, cognitive, emotional, and social outcomes.

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Ethics approval

The procedures used in this study adhere to the tenets of the Declaration of Helsinki. Approval was obtained from the Macquarie University Human Research Ethics Committee (Project ID: 10903; Approval Number: 520231090346872).

Consent to participate

Written informed consent was obtained from all individual participants included in the study.

Consent for publication

The authors affirm that human research participants provided written informed consent for the publication of de-identified data in line with open science requirements.

Preregistration

This study was not preregistered.

Code availability

Not applicable.

CRediT authorship contribution statement

Cassandra L. Crone: Writing – original draft, Visualization, Validation, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Rachel W. Kallen:** Writing – review & editing, Validation, Supervision, Resources.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The dataset generated and analysed during this study is available via Mendeley Data at <https://data.mendeley.com/datasets/svbbn3nrmk/1>

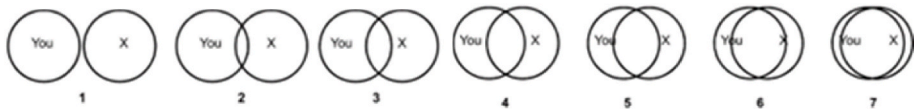
Acknowledgments

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Appendix A

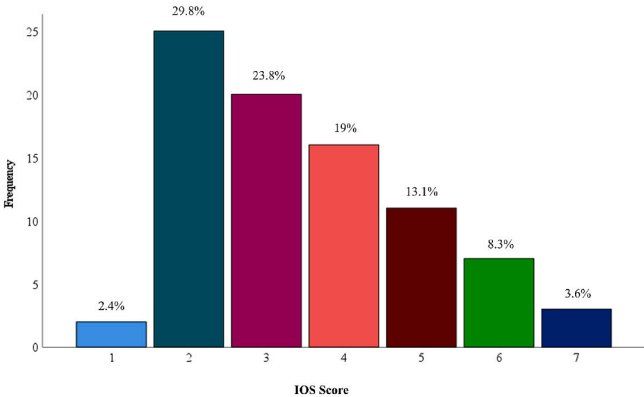
Inclusion of Other in the Self Scale (IOS; Aron et al., 1992)

Which picture best describes the relationship between you and **your avatar**.

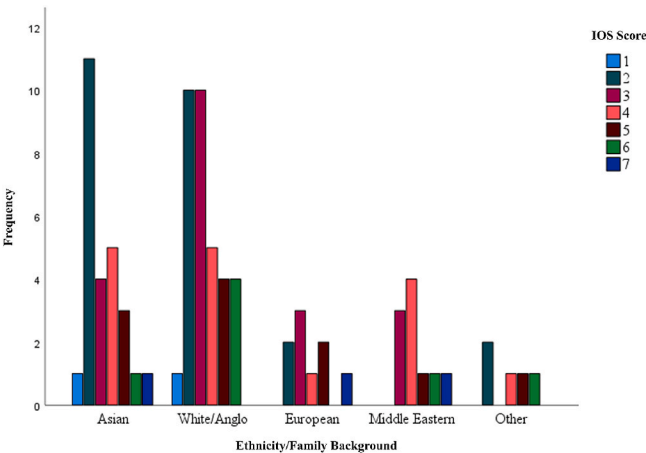


The IOS is a 1-item pictorial measure of perceived closeness between a participant and given target. Participants selected the illustrated degree of perceived overlap between themselves and their assigned avatar. Although specific evaluations of avatar acceptability were beyond the scope of the current study, IOS scores may give some indication of avatar acceptance. IOS scores are displayed below for the full study sample, as well as by participant ethnicity/family background.

Full Sample Frequency of IOS Scores



IOS Scores by Participant Ethnicity/Family Background



Frequencies by Participant Ethnicity/Family Background

Ethnicity/Family Background	1	2	3	4	5	6	7
Asian	3.9%	4.2%	15.4%	19.2%	11.5%	3.9%	3.9%
Anglo/White	2.9%	29.4%	29.4%	14.7%	11.8%	11.8%	0%
European	0%	22.2%	33.3%	11.1%	22.2%	0%	11.1%
Middle Eastern	0%	0%	30%	40%	10%	10%	10%
Other	0%	40%	0%	20%	20%	20%	0%

Note. Other includes Mixed ($n = 3$), Black ($n = 1$), Hispanic ($n = 1$).

Appendix B

Zero-Order Correlations

	1	2	3	4	5	6	7	8	9	10	11
1. EQ_1	–										
2. EQ_2	–0.11	–									
3. EQ_3	0.30	0.28	–								
4. EQ_4	0.68	–0.20	0.30	–							
5. EQ_5	–0.27	0.48	0.16	–0.32	–						
6. EQ_6	0.44	–0.23	0.03	0.50	–0.16	–					
7. EQ_7	0.20	–0.20	–0.08	0.11	0.01	0.54	–				
8. EQ_8	0.30	0.25	0.20	0.10	0.19	0.06	–0.14	–			
9. EQ_9	0.10	0.49	0.39	0.09	0.23	–0.17	–0.23	0.34	–		
10. EQ_14	0.50	–0.20	0.15	0.49	–0.31	0.25	0.15	0.01	0.03	–	
11. EQ_15	0.18	0.46	0.35	0.13	0.23	–0.05	–0.28	0.39	0.37	–0.06	–

Appendix C

Anti-Image Correlations

	1	2	3	4	5	6	7	8	9	10	11
1. EQ_1	0.72 ^a	–									
2. EQ_2	–0.02	0.76 ^a									
3. EQ_3	–0.14	–0.10	0.81 ^a								
4. EQ_4	–0.43	0.16	–0.18	0.69 ^a							
5. EQ_5	0.20	–0.30	–0.16	0.08	0.73 ^a						
6. EQ_6	–0.04	0.02	0.07	–0.41	0.02	0.65 ^a					
7. EQ_7	–0.21	0.02	0.01	0.23	–0.22	–0.53	0.52 ^a				
8. EQ_8	–0.34	0.01	0.06	0.16	–0.18	–0.13	0.18	0.64 ^a			
9. EQ_9	0.04	–0.32	–0.20	–0.13	–0.02	0.13	0.04	–0.20	0.78 ^a		
10. EQ_14	–0.25	0.03	–0.04	–0.20	0.12	0.08	–0.08	0.05	–0.07	0.83 ^a	
11. EQ_15	–0.11	–0.29	–0.13	–0.08	–0.07	–0.06	0.22	–0.18	–0.04	0.10	0.78 ^a

Note. ^a Diagonal elements indicate measures of sampling adequacy.

Anti-Image Covariance

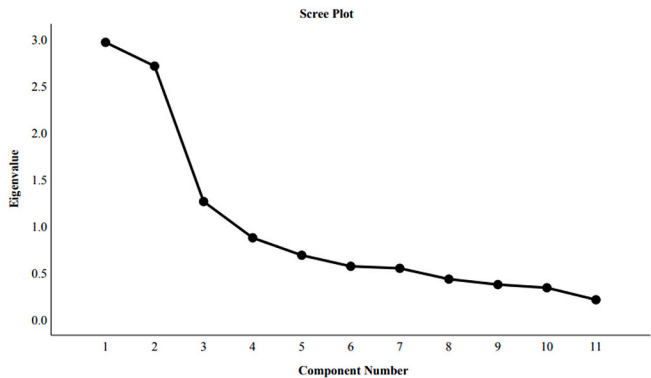
	1	2	3	4	5	6	7	8	9	10	11
1. EQ_1	0.39	–									
2. EQ_2	–0.01	0.53	–								
3. EQ_3	–0.07	–0.06	0.69	–							
4. EQ_4	–0.17	0.07	–0.09	0.38	–						
5. EQ_5	0.10	–0.17	–0.10	0.04	0.62	–					
6. EQ_6	–0.02	0.01	0.04	–0.17	0.01	0.47	–				
7. EQ_7	–0.10	0.01	0.01	0.11	–0.13	–0.27	0.56	–			
8. EQ_8	–0.18	0.01	0.04	0.08	–0.12	–0.07	0.11	0.68	–		
9. EQ_9	0.02	–0.19	–0.13	–0.06	–0.01	0.07	0.02	–0.13	0.63	–	
10. EQ_14	–0.13	0.02	–0.03	–0.10	0.08	0.04	–0.05	0.04	–0.04	0.66	–
11. EQ_15	–0.05	–0.17	–0.09	–0.04	–0.05	–0.03	0.13	–0.12	–0.02	0.06	0.61

Appendix D

Communalities

EQ_1	0.762
EQ_2	0.667
EQ_3	0.462
EQ_4	0.753
EQ_5	0.711
EQ_6	0.758
EQ_7	0.811
EQ_8	0.392
EQ_9	0.551
EQ_14	0.525
EQ_15	0.556

Appendix E



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